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# Application of anisotropic NMR parameters to the confirmation of molecular structure

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**The use of anisotropic NMR data, such as residual dipolar couplings (RDCs) and residual chemical shift anisotropies (RCSAs), has emerged as a powerful technique for structural characterization of organic small molecules. RDCs typically report the relative orientations of different $^{1}$H–$^{13}$C bonds; RCSAs report the relative orientations of different carbon chemical shielding tensors and hence are more useful for proton-deficient molecules. This information is complementary to that obtained from conventional NMR data such as *J* couplings, isotropic chemical shifts, and nuclear Overhauser effects (NOEs)/rotational frame nuclear Overhauser effects (ROEs). Obtaining anisotropic NMR data requires the creation of an anisotropic sample environment through an alignment medium. Here, we focus on the use of compressed or stretched polymeric gels as two different but fundamentally equivalent methods for introducing sample anisotropy. Protocols are provided for the synthesis of the chloroform-compatible poly(methyl methacrylate) and dimethyl sulfoxide (DMSO)-compatible poly(2-hydroxyethyl methacrylate) gels and sample setup with a preparation time of 2–3 d. The bond-specific RDC data and the atom-specific RCSA data are extracted as changes in $^{1}$H–$^{13}$C couplings and $^{13}$C chemical shifts, respectively, between two measurements under different alignment conditions, with a total experimental time of 0.5–4 d. NMR data acquisition and important considerations are described in detail. We also provide step-by-step procedures for the density functional theory (DFT) calculations involved and data analysis using the commercial software MSpin. We use three example compounds, namely cryptospirolepine (505 Da), retrorsine (351 Da), and estrone (270 Da), to demonstrate some important aspects of the workflow, such as input data preparation, handling of structural flexibility, and RCSA data correction when necessary.**

## Introduction

Mischaracterization of natural product structures has been an ongoing problem since the earliest days of organic chemistry, engendering a steady stream of synthetic, computational, and spectroscopic studies to correct the literature and make sense of confounding structure–activity relationships and physical properties[1–4]. Structural misassignments can arise in many ways, but they typically result from misinterpretation of available NMR data or insufficient data to address the specific structural features of the molecule being investigated. An illustrative example is homodimericin A (Fig. 1), for which a standard suite of NMR experiments, including heteronuclear single quantum coherence spectroscopy (HSQC) and heteronuclear multiple bond correlation spectroscopy (HMBC) were insufficient to determine the constitution and configuration[5]. Homodimericin A illustrates how as molecules become progressively more proton deficient, the challenge of determining the molecular constitution correspondingly increases, and even attempts to utilize computer-assisted structure elucidation (CASE) methods[6–8] may fail to give insight into the correct structure of the molecule under study. In the homodimericin case, results of this type are not particularly surprising, considering that the $C_{20}$ hexacyclic central core of the molecule incorporates 14 non-protonated carbons, 11 of them contiguous and 5 of them stereogenic.

In fact, only after incorporation of 1,1-HD-ADEQUATE (homonuclear decoupled adequate sensitivity double-quantum spectroscopy) $^{13}$C–$^{13}$C correlations into the NMR dataset could the constitution of the molecule be elucidated[9]. Indeed, 1,1-ADEQUATE data have also proven useful in

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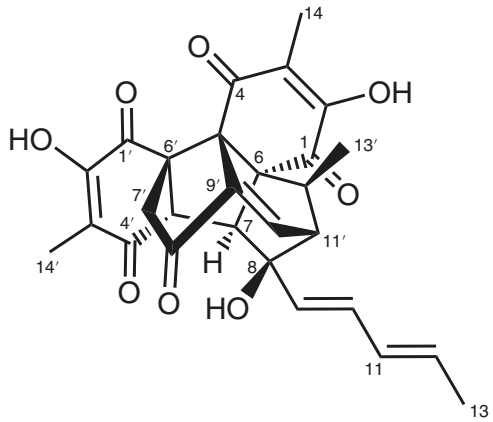



**Fig. 1 |** Structure of homodimericin A.

earlier examples of highly non-protonated scaffolds[10,11]. However, owing to the unprecedented structure of homodimericin A, it was critical to have an independent means of structure validation, as well as to determine the relative configuration of the eight stereogenic centers. Therefore, we adopted an approach that had been used earlier by the Luy group, namely to apply anisotropic NMR parameters to establish the constitution of a molecule[12]. For homodimericin A (Fig. 1), an ensemble of RDC and RCSA data were used to independently confirm the constitution and to determine the molecule's relative configuration. In just over 6 months from our initial report, the structure of homodimericin A was subsequently confirmed by no less than three synthetic studies[13–15].

Structural challenges such as the homodimericin A example above are fostering the rapidly growing interest in the acquisition and utilization of anisotropic NMR data for complex small molecules as an independent source of constraints for structural validation. The use of anisotropic NMR parameters as a general structural elucidation tool began about 2 decades ago in the biomolecular NMR field[16–18], and ~15 years ago in small-molecule NMR with the application of RDCs to structure-elucidation problems[19–35]. More recent work from the group headed by Luy and the authors of this report has made it feasible to measure RCSAs with enough accuracy for routine usage in small-molecule NMR[36–38]. As will be discussed in the 'Origins and manifestation of anisotropic NMR parameters' section, RDC and RCSA are the residual manifestations of the prominent anisotropic NMR interactions in solids, specifically dipolar couplings (DCs) and chemical shift anisotropy (CSA). In an isotropic solution, random molecular tumbling renders both DC and CSA unobservable. By contrast, the anisotropic environment inside a compressed or stretched polymeric gel does not allow fully random molecular rotation, and consequently, a residual level of DC and CSA, usually on the order of 0.1% of the corresponding solid-state value, can be detected. Unlike many other NMR parameters, such as scalar ($J$) couplings and isotropic chemical shifts ($\delta$), the measurement of RDCs and RCSAs requires a control experiment under a different alignment condition in order to be able to eliminate the isotropic contribution to the NMR interactions. For example, a typical RDC measurement is based on the difference in coupling between an aligned sample, which yields the sum of RDC ($D$) and $J$ couplings, and another measurement in isotropic solution, which only yields the $J$ coupling. Unfortunately, taking the chemical shift difference between these two samples does not directly yield RCSA, because the isotropic chemical shift is generally different between such two samples and will not be cleanly removed by taking the difference. Instead, we extract both RDC and RCSA from two measurements on the same sample but with different degrees of compression or stretching, as described in two later sections under the headings 'Residual dipolar couplings' and 'Residual chemical shift anisotropies'.

The two anisotropic techniques are highly complementary. Whereas RDCs provide information about relative angles between internuclear vectors, typically direct C–H bond vectors, RCSAs afford information about relative orientations of chemical shift tensors and hence can report on both protonated and non-protonated carbons in any organic structure. These data can be combined to more robustly evaluate a constitution proposed either by the human investigator or by an artificial intelligence program, usually referred to as a CASE program[39–41]. The choices for CASE programs include CHEMICS[42], the Advanced Chemistry Development Structure Elucidator program[43,44], COCON[45,46], and LSD[47,48]. One additional program, Mnova CASE, has just been released[49]. It should be noted that most CASE programs address chemical constitution problems rather than

configurational ones. The StereoFitter module, being implemented by Mestrelab Research, will incorporate this missing configurational capability and will eventually be fully integrated into the Mnova CASE program, allowing fully automated utilization[50].

To make use of RDC or RCSA data, 3D conformers, i.e., 3D atomic coordinates such as those specified in the Protein Data Bank (PDB) files, must be generated based on proposed molecular constitutions and configurations through computational chemistry methods, such as the DFT, as described in the section on 'Computation of 3D conformations and chemical shielding tensors' below. In the overall structural evaluation, the experimentally measured anisotropic NMR data are compared with the expected values from theoretically calculated geometries and chemical shielding tensors generated by DFT through a fitting procedure[51,52]. Congruence between the experimental and calculated data is evaluated by a quality factor ($Q$ value)[53], which reflects the average percentage disagreement between experimental and calculated RDC or RCSA values, as defined in Eq. 1. Here, $N$ represents the total number of data points measured, and $V_i^{exp}$ and $V_i^{calc}$ represent the experimental and calculated values of the $i$th data point. A low $Q$ value represents good congruence and supports the proposed structure, whereas considerable scatter in the data, accompanied by a high $Q$ value, indicates potential errors with the proposed constitution and/or configuration, and one should evaluate alternative structural proposals that may improve the agreement.

$$Q = \sqrt{\frac{\sum_{i=1}^{N}\left(V_i^{exp} - V_i^{calc}\right)^2}{\sum_{i=1}^{N}\left(V_i^{exp}\right)^2}}. \quad (1)$$

Our goal here is to provide investigators with a practical guide for the acquisition and utilization of anisotropic NMR data for chemical structure elucidation. Key topics discussed will include the preparation of the gels employed as compressed or stretched alignment media, optimization of molecular alignment conditions, the selection of NMR experiments used to acquire the anisotropic NMR data, and finally the computational modeling and the structure verification process. The latter includes conformer generation, geometry optimization, and chemical shielding calculation by DFT methods, followed by anisotropic NMR data analysis based on singular value decomposition (SVD).

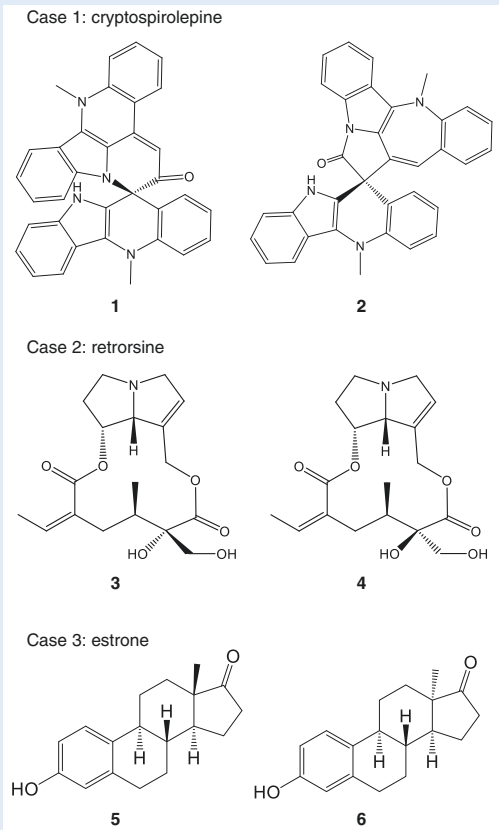
The following three compounds are used as examples to demonstrate the workflow described in this procedure: cryptospirolepine[9] (505 Da), retrorsine[38,54] (351 Da), and estrone[36,38] (Box 1). These illustrate the use of anisotropic NMR data to establish the constitution and configuration of compounds of varying levels of structural plasticity.

## Origins and manifestation of anisotropic NMR parameters

For diamagnetic molecules dissolved in isotropic solvents, anisotropic NMR interactions, such as DCs, CSA, and quadrupolar couplings, average to negligible values due to random molecular tumbling. To recover these interactions for use in solution NMR analyses, it is usually necessary to partially align the molecule being studied relative to the external magnetic field through some sort of alignment medium. For small molecules, such alignment is achieved by either the addition of liquid crystal phase–forming polymers, such as poly-γ-benzyl-L-glutamate (PBLG)[55–57], polyacetylenes[58], and polymer-grafted graphene oxide[59,60]; or by preparing cross-linked polymeric 'gels' that can be swollen with a deuterated solvent solution of the analyte and then mechanically deformed through either compression or stretching[61–69]. More detailed discussion can be found in the 'Compressed or stretched polymeric gels as the alignment medium for small organic molecules' section below. The alignment medium imposes an anisotropic orientation distribution that can be mathematically described in the molecular frame of the molecule by a symmetric and traceless tensor: $A_{ij}$[70]. When diagonalized, the three principal values of the tensor represent the deviation from those of the isotropic orientation distribution, i.e., 1/3 probability each along the three principal axes, $x$, $y$, and $z$. Although the anisotropic components of the NMR parameters average to zero in isotropic solution, this is no longer the case in anisotropic medium, enabling the controlled observation of residual anisotropic NMR parameters, i.e., RDCs, RCSAs, and RQCs (residual quadrupolar coupling), as discussed in the next two sections. The underlying mathematics describing the alignment phenomenon and RDC can be found in a conceptual article by Kramer et al.[70].

**Box 1 | Cryptospirolepine, retrorsine, and estrone as case studies for constitution and configuration analysis**

The structure of cryptospirolepine was originally determined incorrectly as 2 (ref. [105]), based solely on COSY, $^{1}$H–$^{13}$C HMQC, and 8 Hz-optimized $^{1}$H–$^{13}$C HMBC data. It was later revised to structure 1 using the recently developed 1,1-HD-ADEQUATE experiment[9]. The revision was further confirmed by anisotropic NMR data in our recent work[39]. Here, we use data from this compound to illustrate the procedure for RDC- and RCSA-based configuration determination. As a second example, we describe another recent study of ours that used RCSA data for the stereochemical differentiation of retrorsine[38]. Earlier work that used RDC data alone was unable to distinguish the correct structure of retrorsine, 3, from one of its epimers, 4, which involves an inversion of configuration at the C11 quaternary carbon[64]. As a final example, we will utilize the differentiation of estrone, 5, and 13-epi-estrone, 6, to illustrate the isotropic shift correction procedure that is necessary to extract correct RCSA data from compression measurements. Through these case studies we will also demonstrate the procedure of using the single-alignment tensor approach for structural analysis of a molecule of moderate structural flexibility.




## Residual dipolar couplings

For weak coupling conditions, the Hamiltonian operator for the heteronuclear RDC is identical to the Hamiltonian operator of the heteronuclear $J$ coupling in the spin part ($\mathbf{I_z S_z}$), and thus a coupling measurement in an anisotropic medium yields a total coupling $T$ that represents the sum of the scalar coupling and the RDC: $T = J + D$ (scalar ($J$), and dipolar ($D$)). Consequently, the $J$ coupling should be accounted for by comparison of the total coupling values in two different alignment conditions. These conditions can be either a measurement in isotropic and anisotropic conditions or measurement in two anisotropic conditions with different alignment tensors:

$$\mathrm{RDC_{kl}} = T_{\mathrm{kl}}^{(1)} - T_{\mathrm{kl}}^{(2)} = \frac{-3\mu_0 h \gamma_{\mathrm{k}} \gamma_{\mathrm{l}}}{16\pi^3 \left|\overrightarrow{r^{\mathrm{kl}}}\right|^5} \sum_{i,j=x,y,z} \left(A_{ij}^{(1)} - A_{ij}^{(2)}\right) r_i^{\mathrm{kl}} r_j^{\mathrm{kl}}. \quad (2)$$

The constants $\mu_0$ and $h$ represent vacuum permeability and the Planck constant, respectively. The term $RDC_{kl}$ in Eq 2 represents the RDC between nuclei k and l; $T_{\mathrm{kl}}^{(1)}$ and $T_{\mathrm{kl}}^{(2)}$ represent total couplings

under alignment conditions (1) and (2); $\left|\overrightarrow{r^{kl}}\right|$ represents the length of the internuclear vector $\overrightarrow{r^{kl}}$; $r_i^{kl}$ and $r_j^{kl}$ represent the projections of the vector $\overrightarrow{r^{kl}}$ along the $i$ and $j$ axes; and $A_{ij}^{(1)}$ and $A_{ij}^{(2)}$ are the alignment tensor matrix elements under alignment conditions (1) and (2), respectively. Note that the RDC in Eq. 2 in fact stands for the RDC difference between alignment conditions (1) and (2) due to the way it is measured as described above. However, if we simply treat the difference in alignment tensors, i.e., $\widehat{A^{(1)}} - \widehat{A^{(2)}}$, as a third effective alignment tensor, which is also clearly traceless and symmetric because both $\widehat{A^{(1)}}$ and $\widehat{A^{(2)}}$ are, the difference in RDC is simply the RDC associated with this effective alignment tensor and thereby causes no difference in terms of structural analysis. For this reason, we will refer to the RDC difference just as RDC for simplicity. The same simplification also applies to the RCSA below. Note that the conventional method is to first measure $J$ couplings in an isotropic sample and then measure $T$ in an anisotropic sample. However, if RDCs and RCSAs are going to be used simultaneously for structural analysis, RDCs should be measured from the same two degrees of alignment as used for RCSA measurement, so that $\widehat{A^{(1)}} - \widehat{A^{(2)}}$ is consistent with both types of data.

### Residual chemical shift anisotropy

RCSAs have only recently emerged as a general structural tool for small molecules. RCSAs report on the anisotropy of the chemical shielding tensor and its orientation with respect to the alignment frame. Analogous to RDC measurements, chemical shifts in an anisotropic medium are also the sum of the RCSA and the isotropic chemical shift. Accordingly, the RCSA is measured as the difference in chemical shift between two alignment conditions that are defined by the respective alignment tensors $A_{ij}^{(1)}$ and $A_{ij}^{(2)}$.

$$\begin{aligned} \mathrm{RCSA_k} &= \left(\delta^{(1,\mathrm{k})} - \delta^{(1,\mathrm{ref})}\right) - \left(\delta^{(2,\mathrm{k})} - \delta^{(2,\mathrm{ref})}\right) \\ &= \sum_{i,j=x,y,z} \left(A_{ij}^{(1)} - A_{ij}^{(2)}\right)\left(\delta_{ij}^{\mathrm{CSA_k}} - \delta_{ij}^{\mathrm{CSA_{ref}}}\right). \end{aligned} \tag{3}$$

where $\mathrm{RCSA_k}$ represents the RCSA of nucleus k; $\delta^{(1,\mathrm{k})}$, $\delta^{(1,\mathrm{ref})}$, $\delta^{(2,\mathrm{k})}$, and $\delta^{(2,\mathrm{ref})}$ represent the chemical shifts of nucleus k and a reference nucleus under alignment conditions (1) and (2); and $\delta_{ij}^{\mathrm{CSA_k}}$ and $\delta_{ij}^{\mathrm{CSA_{ref}}}$ represent the CSA tensor elements of nucleus k and the reference nucleus. An arbitrary carbon atom in the molecule can be selected as the reference so that RCSA measurement becomes independent of chemical shift calibration[36]. Tetramethylsilane (TMS) can also be used as a RCSA reference compound. Owing to the tetrahedral molecular geometry, TMS has no RCSA at the equilibrium geometry and aligns much more weakly than most compounds of interest, although vibration–rotation coupling can generate measurable anisotropic effects in tetrahedral molecules under very strong alignment conditions[71–73]. Therefore, the TMS methyl carbon RCSA is expected to be negligible in comparison to RCSAs of a typical compound under an identical alignment condition. If TMS is used for carbon referencing, Eq. 3 further simplifies to Eq. 4 below, as shown previously[38].

$$\mathrm{RCSA_k} = \sum_{i,j=x,y,z} \left(A_{ij}^{(1)} - A_{ij}^{(2)}\right)\delta_{ij}^{\mathrm{CSA_k}}. \tag{4}$$

When changing the degree of alignment, e.g., by increasing the stretching or compression ratio of the swollen gel, uncontrollable variations in the isotropic chemical shifts can easily be introduced, making it extremely difficult to measure the RCSAs with sufficient accuracy. After initial attempts by Luy and coworkers[36,37], we recently developed a gel stretching method that makes the isotropic shift variation negligible, as well as a gel compression method that allows correction for the isotropic shift variation; both methods utilize routinely available NMR instrumentation, thereby providing easier access to these useful parameters[38]. Further details about these methods can be found in a later section under the heading 'Properties of the compression and the stretching methods'.

### Combining RDCs and RCSAs for structural analysis

As shown in Eqs. 2 and 3, RDCs and RCSAs are related through the same alignment tensor and can therefore be used simultaneously to analyze the orientations of C–H bonds relative to chemical shielding tensors. Such information is not available from using either RDCs or RCSAs alone; using the two techniques in tandem provides a far more robust structural analysis with a diminished possibility of structural misassignments, especially for many small molecules with limited data availability.

When RCSA and RDC data are simultaneously employed in structural analysis, it is important to weight them properly to reflect their relative significance. The size of RCSA measurement, when measured in hertz, increases linearly with respect to the field strength. By contrast, RDCs, when also expressed in hertz, are field independent. The theoretical ratio between the data spans of RCSA and RDCs, i.e., the difference between the smallest value and the largest value among randomly orientated CSA tensors or bond vectors in the molecular frame, is described by[74]:

$$R = \frac{32\pi^3 \left|\overrightarrow{r^{kl}}\right| \nu_0(\delta_{11} - \delta_{33})}{9\mu_0 h \gamma_k \gamma_l}, \tag{5}$$

where the constants $\mu_0$ and $h$ represent vacuum permeability and the Planck constant, respectively, $\nu_0$ is the Larmor frequency of the atom for which the RCSA is measured, and $\delta_{11}$ and $\delta_{33}$ are the largest and smallest eigenvalues of the chemical shift tensor. For example, at commonly utilized NMR observation frequencies from 500 to 900 MHz, the amide carbonyl RCSA is ~30–60% of the $^{13}C$–$^{1}H$ RDC in amplitude. If similar measurement errors are assumed for RDCs and RCSAs, which is reasonable if the RDC is measured from the F1 (carbon) dimension (see 'Experimental measurement of anisotropic NMR data'), a straightforward way of weighting RDC and RCSA data is simply expressing both in hertz, which reflects the fact that the relative significance of RCSAs increases with the magnetic field strength.

It is also worth noting that the carbon CSA critically depends on hybridization. An $sp^2$ carbon generally has a larger CSA than an $sp^3$ carbon and therefore leads to a larger RCSA at the same CSA tensor orientation in the molecular frame. For this reason, RCSAs are expected to be more useful in compounds with $sp^2$ carbons, as their values can be measured with higher accuracy, in addition to the fact that their CSA tensors can be usually calculated with lower relative errors by DFT.

### Compressed or stretched polymeric gels as the alignment media for small organic molecules

Small organic molecules can be oriented in two types of alignment media compatible with organic solvents: (i) compressed or stretched cross-linked polymeric gels swollen in organic solvents and (ii) lyotropic liquid crystalline solutions of helical polymers, including polypeptides[55–57,75–77], other polymers[58,78,79], or polymer-grafted graphene oxide[59,60]. The present protocol is focused exclusively on the utilization of alignment of polymer gels, also known as strain-induced alignment in gels (SAG), for the measurement of both RDCs and RCSAs. NMR work on strained gel polymers was first reported in 1981 when Deloche and Samulski[80] studied rubber elasticity by measuring the deuterium quadrupolar coupling of the solvent as a function of gel straining. On the basis of this work, Tycko et al.[81], as well as Grzesiek and coworkers[82], independently reported in 2000 the first utilization of the SAG method to align proteins in anisotropically compressed polyacrylamide (PAAM) gels swollen in water. A disadvantage is that the dense gel matrix used in these studies hinders rotational diffusion, causing undesirable line broadening and consequently decreased sensitivity[18]. This problem was alleviated by stretching the gels in the direction parallel to the applied magnetic field. Axially stretching the gel causes the long axis of rod-shaped proteins to align in parallel, instead of orthogonally, to the axis of the magnetic field, doubling the size of RDCs obtained for the same gel density, so that a more dilute gel can be employed for a better line shape[83]. The stretching device proposed by Chou et al.[83] for aqueous samples is commercially available and widely used in the application of RDCs in biomolecular NMR.

The SAG method has also been widely applied in small-molecule NMR over the past 15 years, with the development of various polymer gels that swell in different organic solvents (indicated in parentheses), including the following.

- Polystyrene (PS) gel (chloroform) and polyacrylonitrile (PAN) gel (DMSO), used by Luy and coworkers[61,66].
- Polyacrylamide-based copolymer gel (DMSO), used by Griesinger and coworkers[62].
- Poly(methyl methacrylate) (PMMA) gel (chloroform/dichloromethane)[63], poly(2-hydroxyethyl methacrylate) (poly-HEMA) gel (DMSO)[64], and di(ethylene glycol) methyl ether methacrylate (poly-DEGMEMA) gel (methanol)[65] are examples of other gels developed by Gil and coworkers that have found widespread application.
- Polyethylene oxide (PEO) gel (developed by Luy and coworkers) is the most versatile gel to date in terms of solvent compatibility, but unfortunately, preparation of this gel requires cross-linking with γ-irradiation, which is not readily available to most researchers[84].

In the SAG method, the gel must be either compressed or stretched in order to generate an anisotropic medium environment for the alignment of the analyte molecule. Both compression and stretching methods were used in the early 2000s to align biomolecules, along with the development of SAG[81–83]. A reversible compression technique was later developed by Gil and coworkers in 2010 for small-molecule alignment through the design of a screw-based compression device[69], which will be discussed further later in this protocol. Most of the early applications of SAG in small molecules were based on the self-stretching of a gel in a regular NMR tube as the gel absorbs the solvent and elongates[61,62,66]. The self-stretching process is typically lengthy, and the alignment condition cannot be adjusted with the same gel. In 2006, Kuchel et al.[67] reported a device for adjustable gel stretching that they applied to a gelatin gel for enantiomer discrimination using RDCs and $^{23}$Na quadrupolar couplings. Luy et al. extended the application of Kuchel's apparatus to all gel/solvent combinations[68] by adapting a flexible, perfluorinated elastomer tube that is resistant to most organic solvents. Finally, the glass-based gel stretching device, also used in this study, was initially reported in 2010 by Liu and Prestegard[74] for the measurement of RDCs and RCSAs in soluble and membrane-associated proteins. This stretching device was recently adapted to measure RDCs and RCSAs of small molecules in organic solvents[5,38,39,41]. Both devices described in this protocol have been used with PMMA and poly-HEMA gels, and could most likely be used with the other aforementioned gels as well.

## Properties of the compression and the stretching methods

The following introduction covers features of the gel compression method using the screw-based compression device by Gil and coworkers[24,69], and the gel stretching method using the glass-based stretching tube described by Liu and Prestegard[5,38,39,74].

### Reversible compression/relaxation method

Compression is introduced using the commercially available New Era device. When the position of the piston is locked at a given degree of compression, the nut and bolt seal the tube tightly and long-duration experiments can be undertaken with negligible $CDCl_3$ evaporation. Gels can be compressed to generate any degree of alignment between nearly zero and the maximum by adjustment of the piston position in a reversible fashion. Because the swollen gel normally has a diameter of ~3.6–3.8 mm, and the inner diameter of a 5-mm tube is typically ~4.0–4.2 mm, there is always a layer or pockets of isotropic solution present between the gel and the wall of the tube, even when the gel is fully compressed[85]. As a result, signals from the solution inside and outside the gel are simultaneously observed in NMR spectra. Although seemingly an impediment, the isotropic solution can be used to an advantage, allowing scalar couplings ($J$) and total couplings ($T = J + D$) to be collected simultaneously[86]. RCSA data collected with the compression method must be corrected before analysis because of changes in polymer/solvent ratio[38]. Briefly, as the gel is compressed, some liquid is also expelled from the gel matrix, decreasing the gel solvation and consequently causing a change in the isotropic component of chemical shielding. This isotropic component change coexists with the alignment-induced frequency change due to the anisotropic components of chemical shielding, i.e., RCSA, and must be eliminated. By assuming a constant relationship between the isotropic chemical shift change and the variation in gel solvation upon compression, the isotropic component of the chemical shift can be removed post acquisition before data analysis[38]. A recently published algorithm can remove the isotropic chemical shift interference automatically during the SVD fitting in MSpin (see 'Anisotropic NMR data analysis' below) without the need for any additional measurement[86].

### Stretching method

As previously described, the gel used in the compression device is in a fully swollen state during NMR data acquisition. By contrast, the gel in the stretching device is only partially swollen and thereby 'solvent-thirsty', i.e., with sufficient solvent, the gel will expand to a substantially greater size than it is allowed to adopt during NMR measurements. As a result, all liquid is contained in the gel in the stretching apparatus, and consequently, NMR signals arising from the analyte present in the isotropic interstitial liquids, normally observed in compressed gels, are absent, yielding cleaner NMR spectra without signals from isotropic solution. Furthermore, because the solution is always contained inside the gel matrix during both the weak and strong alignment conditions, gel solvation remains constant, and therefore RCSA measurement is not affected by changes in isotropic chemical shifts. Consequently, RCSA data extracted from the chemical shift differences measured for the weakly and strongly stretched gel can be used directly after proper chemical shift referencing, eliminating the

need for any isotropic shift correction and also avoiding any potential errors propagated through the correction procedure. Conversely, stretching a gel results in a reduction of sample mass in the probe radiofrequency (rf) coil volume, which correspondingly lowers sensitivity. For this reason, more scans, typically two to fourfold, should be accumulated for the strongly stretched sample relative to the weakly stretched condition, so that comparable signal-to-noise ratios are obtained in the two measurements. The mass reduction can be alleviated by using a more heavily cross-linked gel in conjunction with more moderate stretching by adopting a narrow section with a larger inner diameter. Because the stretching device has two open ends that must be sealed with rubber stoppers during the NMR data acquisition, evaporation of volatile $CDCl_3$ around the sealing stoppers during long experiments can cause gel shrinkage that degrades sample shimming and leads to changes in sample concentration. Therefore, RDCs and RCSA measurements in $CDCl_3$ should generally be completed within ~12 h once the PMMA gel is transferred to the stretching device, which is typically achievable for a mass of ≥3 mg for small molecules in the range of 300–600 Da. For the DMSO-compatible poly-HEMA gel, evaporation is not an issue, and long acquisitions can be used for signal averaging with low-mass samples. Such differences in sample mass requirements should be considered whenever the compound of interest is soluble in both chloroform and DMSO.

### Tuning the alignment

Compounds with different intrinsic molecular shapes have varying propensities to align in a gel. For example, planar or elongated molecules align more readily than spherical molecules if the steric effect dominates the alignment mechanism. When the alignment is too weak, both RDC and RCSA values will be small, which consequently decreases measurement accuracy. Ultimately, weak alignment can lead to poorer $Q$ values when the data are analyzed. Conversely, if the alignment is too strong, several factors handicap RDC measurement. First, proton resonances are broadened due to homonuclear DCs, which greatly decreases magnetization transfer efficiency, and strong coupling effects are more pronounced. Second, the effective $^{13}C$–$^{1}H$ coupling, $T$, i.e., the sum of the RDC ($D$) and scalar ($J$) coupling, has a large dynamic range, leading to inefficient INEPT (insensitive nuclei enhanced by polarization transfer)-based magnetization transfers. Finally, $T_2$ relaxation times are shortened, leading to decreased sensitivity of HSQC-type experiments. By contrast, RCSAs are affected by strong alignment to a much smaller extent, first, because inhomogeneous line broadening due to unresolved heteronuclear DCs and homogeneous broadening due to proton (A) spin diffusion occurring in an $A_nX$ dipolar coupled system are alleviated by proton decoupling when $^{13}C$ data are acquired, and, second, because no INEPT-based magnetization transfer is needed for the RCSA measurement. In fact, RCSA measurement generally benefits from stronger alignment due to higher measurement accuracy[87]. However, when RDC data are also required, a compromise should be made by tuning the alignment amplitude to constrain the RDC data within a suitable range for optimal spectral quality[54]. Typically, a range of ~30–50 Hz, i.e., the difference between the largest and smallest RDCs in a molecule, is suitable for $^{13}C$–$^{1}H$ RDC measurement, although both smaller and larger RDC ranges have been successfully reported in the literature. For the reversible compression/relaxation method, the alignment amplitude can be tuned anywhere from zero to the maximum by adjusting the piston position; for the glass-based stretching method, readjusting the alignment requires recovering the sample and repeating NMR measurements in a stretching tube with a different stretching ratio, or soaking the analyte in a gel with a different degree of cross-linking ratio or polymer concentration. See Boxes 2 and 3 for details.

### Experimental measurement of anisotropic NMR data

#### RDC data

The most common option for the measurement of one-bond $^{1}H$–$^{13}C$ RDCs ($^{1}D_{CH}$) is the use of $^{1}H$–$^{13}C$ HSQC-based experiments. The simplest realization would be to turn off carbon decoupling during acquisition, thereby furnishing splittings in the F2 direct dimension. To prevent phase distortion of multiplets due to mismatching of the INEPT transfer, the undesired antiphase magnetization should be converted to unobservable multiple-quantum coherence through the use of the CLIP–HSQC experiment[88]. However, for proton-rich molecules, homonuclear proton–proton DCs can severely broaden the observed signals, thereby hampering accurate measurements. Furthermore, for molecules with coupled protons having poor signal dispersion, strong homonuclear coupling may severely affect the measurement[89]. Care should therefore be taken in the measurement of individual couplings in methylene groups, where strong coupling frequently occurs.

Another option is to measure couplings in the indirect dimension while avoiding heteronuclear $J$ refocusing during the $t_1$ period. The lack of accuracy due to poor digital resolution can be substantially alleviated by one of the following three approaches: (i) aliasing the spectrum in F1, thereby allowing the use of a smaller spectral window; (ii) adopting the 2D $J$-resolved HSQC experiment[90,91], which greatly reduces the spectral window (a 500-Hz window suffices in most cases) but may lead to signal overlapping when the proton dimension is crowded; (iii) using the $J$-scaled HSQC experiment in which the apparent $J$ coupling is scaled up by several fold (typically three to four) so that the measurement is less prone to under-digitization[91]. In these experiments, a BIRD (bilinear rotation decoupling) module[92] can be incorporated into the $J$-evolution period to refocus long-range $^{13}C$–$^{1}H$ couplings for sharper signals[91,93]. The sensitivity of these experiments can generally be enhanced by adopting the real-time pure-shift acquisition scheme[94], as implemented in the HD-$J$-HSQC experiment[90]. This type of experiment does not usually allow the individual measurement of couplings in diastereotopic methylene groups but yields only the sum of the two $^{1}D_{CH}$ values, which has an advantage in that the sum is not affected by the potentially strong coupling between the diastereotopic protons. Furthermore, the use of the sum avoids the need for the specific assignment of the diastereotopic protons. In summary, the F2-coupled CLIP–HSQC experiment can be safely used for weakly coupled systems, particularly if individual $^{1}D_{CH}$ data in methylene groups are going to be measured. In other cases, an F1-based experiment should be used.

**RCSA data**

Carbon RCSA values are simply extracted as chemical shift differences between weak and strong alignment conditions. The NMR experiment is a simple $^{13}C$ direct observation experiment with heteronuclear NOE enhancement and broadband proton decoupling, i.e., the 1D $^{13}C\{^{1}H\}$ NMR experiment. Data from the stretched gel can be used directly for structural analysis after proper referencing, either to TMS using Eq. 4 or to an arbitrary carbon atom within the molecule using Eq. 3. Data from the compressed gel, by contrast, must be corrected to compensate for isotropic chemical shift changes due to the shrinking of gel volume under strong compression as described earlier[38,86].

**Computational modeling**

*Conformer search and geometry optimization*. A key step of RDC/RCSA-based structural analysis is to propose all possible scenarios of constitution and configuration for the compound of interest, which are either deduced by human investigators or generated with the aid of a CASE program[39–41]. To apply RDCs and RCSAs to the differentiation of these candidates, realistic 3D conformational models must be built on the basis of the corresponding constitution and configuration of each candidate. This process begins from a molecular mechanics (MM)–based conformer search conducted on each candidate with defined constitution and configuration, the goal being to sufficiently sample the possible conformational space. Next, a higher level of geometry optimization by DFT is performed for a selected set of conformers to obtain greater accuracy on 3D molecular structures as well as the topology of the potential surface and relative conformational energies.

Many commercial software packages are available for MM-based conformer generation. We used the Maestro Schrodinger package here, making use of its MMFF94 force field implementation[95] which provides accurate geometries and adequate treatment of numerous functional groups. We employed the Monte Carlo multiple minimum algorithm[96], and structures were kept in an energy window of 21 kJ/mol. When only RDCs are used, MM-derived geometries are frequently sufficiently accurate to solve many problems[40]. Higher molecular geometry accuracy, as well as the representation of potential surface topology, can be obtained by reoptimizing the structures at a chosen DFT level according to the desired accuracy and available computational resources. We used here a standard level of theory: B3LYP/6-31G**, which uses the DFT–HF (Hartree–Fock) hybrid functional of B3LYP (Becke, three-parameter, Lee–Yang–Parr), and a split-valence double-zeta basis set 6-31G**, in which the core orbital is described by a single-basis function consisting of six Gaussian-type orbitals (GTOs) and each valence orbital is described by two basis functions (double-zeta)—one consisting of three GTOs, the other of one GTO. The two asterisks in the basis set, sometimes also written as ($d$,$p$), indicate adding $d$ polarization to non-hydrogen atoms and $p$ polarization to hydrogens. This level of theory provides good molecular geometry information at a moderate computational cost.

Furthermore, vibrational frequencies should be computed to obtain thermochemical parameters and to differentiate real minima from saddle points in the energy landscape. Transition-state structures with imaginary vibrational frequencies must be discarded. Usually, the DFT geometry optimization and frequency calculation are the rate-limiting steps in the modeling process. The

calculation time required depends on multiple factors, such as the molecular size, the level of structural flexibility, and computer hardware. On a computer cluster or network using Cray architecture, parallel computing with 16 CPUs per submission can typically be finished within a few hours for most small molecules. Because different initial conformers frequently converge to the same stationary points after DFT optimization, redundant structures must be discarded. Depending on the flexibility of the compound in question, the representative conformers may have varying levels of energy separation. A rigid compound will generally be associated with a large energy gap, e.g., >2 kcal/mol, between the lowest and second lowest-energy conformers, whereas several conformations of comparable energy will generally be found for molecules of increasing flexibility. For molecules of modest flexibility, anisotropic NMR data analysis frequently applies a single-alignment tensor approximation to an ensemble of the lowest-energy conformers after proper structural superposition[73,97,98], as will be illustrated in the 'Anisotropic NMR data analysis by MSpin' section.

*Chemical shielding tensor calculation by DFT*. Before RCSA-based structural analysis, chemical shielding tensors of carbon atoms must be calculated for the geometry-optimized conformers using gauge-invariant atomic orbital (GIAO)–DFT. In most cases, this calculation needs to be undertaken only for those conformers of each candidate whose thermal free energies are within a 2 kcal/mol cutoff relative to that of the lowest. The GIAO[99]/MPW1PW91/6-31G**[100] level of theory was used for the work described in this report. In comparison to equilibrium geometry optimization, chemical shielding calculation is much faster, taking only minutes. The performance of various combinations of functionals and basis sets for the determination of isotropic $^{13}$C chemical shifts has been extensively evaluated in the literature[101,102], and we can safely assume that their performance can be extrapolated to the computation of shielding anisotropy. Calculations were performed in vacuo, as we observed that inclusion of implicit solvation minimally affects the quality factors ($Q$) measured.

*Anisotropic NMR data analysis*. The proposed structure is evaluated by comparing the measured RDC and RCSA data with data back-calculated from DFT-calculated geometries and shielding tensors for the proposed structure based on Eqs. 2 and 3. The back-calculation requires determination of the molecular alignment tensor, i.e., the matrix $\hat{A}$ in Eqs. 2 and 3, by solving a set of overdetermined linear equations in the least-square sense through the SVD method[51]. With both measured and back-calculated RDC and RCSA values in hand, a $Q$ factor can be readily calculated by Eq. 1. The same process should be repeated for all possible structural proposals, and the correct structure is expected to have the lowest $Q$ factor. The analysis is straightforward if the compound of interest has only a single dominant conformation, but can be more involved with increasing molecular flexibility.

Internal molecular motions on the picosecond-to-millisecond time scale can reduce the magnitude of RDC and RCSA from the values given by Eqs. 2 and 3. Such motions include the picosecond-time-scale atomic vibrations around an energy minimum, as well as the larger-amplitude exchange motions between different energy minima that occur on the micro-to-millisecond time scale. The bond vector or CSA tensor–specific order parameters could be included in Eqs. 2 and 3 as a scaling factor[70]. These order parameters are often used to depict structural dynamics in biomolecules, for which conformational distribution is not directly accessible. If order parameters are not explicitly used, as is usually the case in small-molecule analysis, the isotropic component of the low-amplitude vibrational dynamics is absorbed into the axial component of the alignment tensor during SVD and is inconsequential for structural analysis. The anisotropic component due to site-specific vibrational differences generally causes minor effects and could be explicitly taken into account through the use of molecular dynamics approaches, if needed[103]. Large-amplitude conformational jumps between different energy minima, which have a much more substantial impact on RDC/RCSA data interpretation than atomic vibrations, can be taken into account by the single-tensor ensemble fitting approach[73,97,98,104]. SVD equations have been extended in MSpin for the management of multi-conformational problems through the single-tensor approximation. This approximation is known to work well for molecules with limited flexibility. However, caution should be taken when working with systems in which the global shape of the molecule changes dramatically between conformations[29]. When the single-tensor approximation is applicable, the user should also critically assess the reliability of DFT energy calculations for the determination of conformational amplitudes, i.e., the populations of different lowest-energy conformers. Modern DFT calculations generally provide good relative energies, and the Boltzmann weighting information can be used in the structure analysis. However, relative energies in systems presenting specific interactions, such as hydrogen bonds, can be difficult to obtain even at a semiquantitative level. In such cases, conformational deconvolution must be performed. MSpin allows the simultaneous optimization of molecular amplitudes during alignment tensor determination, through a Levenberg–Marquardt nonlinear optimization procedure[97].

In some cases, the *Q* factors between different constitutional or stereochemical proposals are similar in size, making the distinction unclear. Changing the alignment medium can be a powerful method for acquiring additional structure data. For example, the alignment tensor of estrone in the chloroform-compatible PMMA gel is linearly independent of that in the DMSO-compatible (*S*)-2-acrylamido-1–propanesulfonic acid (APS) gel[36,38], and consequently, anisotropic data from the two gels are complementary. The differentiation of estrone and 13-*epi*-estrone was much more readily achieved in the PMMA gel than in the APS gel, because of a favorable alignment tensor orientation in the former case. In general, it cannot be reliably predicted which gel will provide the best differentiation for a given problem, so trials of different conditions can be helpful for especially challenging problems.

From a more general point of view, and also for opening future perspectives, we should emphasize that it is also possible to radically increase the amount of information, and therefore discrimination power by combining isotropic and anisotropic data. In this respect, a procedure, implemented in the StereoFitter Mnova modules, was recently described in which RDC data were used jointly with isotropic $^{13}$C and $^{1}$H chemical shifts and scalar couplings for computer-assisted automated relative configuration elucidation of natural products[50]. RDC and RCSA data can also be combined with CASE program output derived from isotropic NMR data in a similar fashion[39].

## Materials

### Reagents

- Methyl methacrylate (MMA; Sigma-Aldrich, cat. no. M55909)
- 2-Hydroxyethyl methacrylate (HEMA; Sigma-Aldrich, cat. no. 128635)
- Ethylene glycol dimethacrylate (EGDMA; Sigma-Aldrich, cat. no. 335681)
- 2,2′-Azobis(2,4-dimethyl-4-methoxyvaleronitrile) (Wako Chemical, cat. no. V-70)
- 2,2′-Azobis(2-methylpropionitrile) (AIBN; Sigma-Aldrich, cat. no. 441090)
- Aluminum oxide (Sigma-Aldrich, cat. no. 1010971000)
- Acetone (Fisher Scientific, cat. no. A11-4)
- Methanol (Fisher Scientific, cat. no. A412-4)
- Chloroform (Fisher Scientific, cat. no. C298-4)
- Dichlorodimethylsilane (≥99.5% (wt/wt); Sigma-Aldrich, cat. no. 440272)
- Deuterated NMR solvents, including $CDCl_3$ and DMSO-$d_6$ (Cambridge Isotope Laboratories, cat. nos. DLM-7-100 and DLM-10-100, respectively)
- Cryptospirolepine (**2**; the alkaloid was isolated chromatographically from the ground roots of the West African plant *Cryptolepis sanguinolenta*, as previously described[105]. A voucher sample of the alkaloid, with a relative purity of ~90%, was used for this study.
- Retrorsine (≥90% (wt/wt); Sigma-Aldrich,cat. no. R0382)
- Estrone (≥99% (wt/wt); Sigma-Aldrich, cat. no. E9750)

### Equipment

- Gel compression and stretching devices were both obtained from New Era Enterprises. The compression device 5-mm tube assembly used in this report was cat. no. NE-375-5. The specific stretched gel tube used in this report was a 5-mm tube with a 4.2-mm i.d. that tapered down to a 3.2-mm i.d. (cat. no. NE-HP5-3.2-GT-7). Other wide/narrow ID-tapered gel tube configurations are also available from New Era Enterprises.
- The microstoppers used with the stretched gel tube were ordered from Sigma-Aldrich (cat. no. Z165204).
- The piston used in the stretching device to position the swollen gel is part of the Shigemi solvent–matched NMR tube assembly (Sigma-Aldrich, cat. no. Z529370). However, solvent matching of the piston to the solvent is unimportant, as long the gel is long enough to fill the entire probe coil space.
- The 1/8-inch-i.d. FEP (fluorinated ethylene propylene) tubing used during the polymerization of poly-HEMA gels for use with the stretching device was procured from Cole-Parmer (cat. no. EW-06450-05).
- The 3-mm-o.d. NMR tubes used for the polymerization of PMMA and poly-HEMA gels for use with the compression device were ordered from New Era Enterprises (cat. no. NE-H3-7).
- The 4-mm-o.d. NMR tubes used for the polymerization of PMMA gels for use with the stretching device were ordered from Wilmad-LabGlass (cat. no. 405-PS-8).
- Parafilm M (Fisher Scientific, cat. no. 13-374-12)
- Hand centrifuge (Humboldt, cat. no. H-9405)

- Bruker NMR spectrometers operating at 500 and 600 MHz for $^1$H observation, both equipped with liquid $N_2$ cooled cryoprobes (Prodigy), were used for the acquisition of the NMR data for cryptospirolepine[105] (**1**) (3 mg) and retrorsine (**3**) (3 mg), using the gel stretching device. The estrone (**5**) data were collected with 2 mg of material in a gel compression device on a 900-MHz Bruker NMR spectrometer equipped with a 5-mm helium-cooled cryoprobe[38].
- Conformational searching was performed with the Maestro Schrodinger package[95] (https://www.schrodinger.com/maestro). Gaussian 09 software was used for all DFT calculations[106] (all our procedures are compatible with the latest version of Gaussian 16 (http://gaussian.com/gaussian16)). Anisotropic NMR data were analyzed using the SVD-based methodology as implemented in MSpin[52] (http://mestrelab.com/download/mspin/).

## Reagent setup

### Preparation of PMMA gels for use with the compression device

1 Pass MMA and EGDMA through a short column packed with aluminum oxide to remove the polymerization inhibitor.
2 Prepare solution A, which contains 10 mL of MMA, 0.003 g of V-70, and 2 mL of acetone.
3 Prepare solution B by mixing 10 mL of solution A with 40 μL of EGDMA. The mixture contains $7.79 \times 10^{-2}$ mol MMA and $2.12 \times 10^{-4}$ mol EDGMA, corresponding to a cross-linking ratio of 0.27 mol%.
4 Transfer solution B to 3-mm-o.d. NMR tubes, cap the tubes with rubber microstoppers, and secure the stoppers to the tube openings with tape. Ten milliliters of solution B will fill approximately 12 NMR tubes.
5 Place the sealed NMR tubes in a 50 °C oil bath or oven for 5 h until polymerization is complete.
6 Remove the tubes, remove the microstoppers, and leave the gels to dry at ambient conditions (room temperature, ~25 °C).
7 Gently break the NMR tubes with a mortar, retrieve the dried gel rods, and cut them into 2.5-cm sticks.
8 Prewash the sticks in a 1:1 (vol/vol) acetone/methanol solution in a glass vial for 8 h.
9 Transfer the prewashed sticks to another glass vial filled with deuterated chloroform, incubate for at least 3 h at ambient temperature, slowly decant the chloroform, and refill the vial with fresh deuterated chloroform. Repeat this step three times to thoroughly wash the gels.
10 Remove the gels from the glass vial and leave them to dry on a glass surface inside the fume hood for 2 h. The dried gel sticks can be used for NMR or kept for long-term storage (up to 2 years).

### Poly-HEMA gels for use with the compression device

These are prepared following the same steps as for the PMMA gels described above, except for the following differences:

1 The 3-mm-o.d. NMR tubes are silanized before use. Fill the tubes with dichlorodimethylsilane in a fume hood and silanize for ~2 h. Wear gloves, a lab coat, and eye protection when working with this silanizing agent. Decant the silanizing solution back into the storage bottle. Dry the tubes with a nitrogen stream. Leave the tubes in a fume hood for ~2 h, after which the tubes are ready to be used.
2 MMA is replaced by HEMA in solution A.
3 Acetone is replaced by an equal volume of methanol in solution A.
4 Deoxygenate solution B by sonication for 10 min in a closed container, followed by bubbling a slow and steady nitrogen stream through the liquid for 1 min.
5 Prewashing is not needed for poly-HEMA gels.
6 Use methanol instead of chloroform to wash the poly-HEMA gels.

### PMMA gels for use with the stretching device

These are prepared in a way similar to that for PMMA gels prepared for use with the compression device, although somewhat different steps and conditions are taken and used. The procedure for making the 70% (vol/vol) PMMA gel with 0.01% (mol/mol) EDGMA cross-linking is given below. The recipe can be easily modified to prepare PMMA gels of other concentrations and cross-linking ratios.

1 Purify MMA and EGDMA with an aluminum oxide column.
2 Prepare a 100× diluted EGDMA stock solution (1% vol/vol EGDMA in MMA) by mixing EGDMA and MMA at 1:99 (vol/vol) ratio.
3 Prepare a 6% (wt/vol) AIBN stock solution in methanol. Vortex to dissolve the AIBN completely. The stock solution can be frozen at −20 °C (for up to 6 months) for later use.
4 Prepare the polymerization mixture solution of 10-mL total volume by mixing the following components: 7 mL of MMA, 125 μL of 1% (vol/vol) EGDMA in MMA, 50 μL of 6% (wt/vol) AIBN, and 3 mL of acetone.

5 Transfer the mixture to 4-mL-o.d. NMR tubes, cap the tubes, and secure them with Parafilm M. Ten milliliters of the solution will fill approximately seven tubes.
6 Incubate the tubes in a 65 °C oven overnight until polymerization is complete.
7 Carefully break the NMR tubes with a mortar, retrieve the dry gel rods, and cut them into 2-cm sticks.
8 The prewash, wash, and the final drying steps are identical to those used for PMMA gels for use with the compression device.

**Poly-HEMA gels for use with the stretching device**

The steps for making the 60% (vol/vol) poly-HEMA gel with 0.04% (mol/mol) EDGMA cross-linking are listed below. The recipe can be easily modified to prepare poly-HEMA gels of other concentrations and cross-linking ratios.

1 Cut the 1/8-inch-i.d. FEP tubing into ~30-cm segments, and securely plug one end of each segment with a microstopper.
2 Purify HEMA and EDGMA by passing them through an aluminum oxide column.
3 Prepare a 100× diluted EGDMA stock solution (1% vol/vol EGDMA in HEMA) by mixing EGDMA and HEMA at a 1:99 (vol/vol) ratio.
4 Prepare a 6% (wt/vol) V-70 stock solution in acetone. Vortex the mixture to dissolve the V-70 completely. The stock can be frozen at −20 °C for up to 6 months for later use.
5 Prepare the polymerization mixture solution of ~10-mL total volume by mixing the following components: 6 mL of HEMA, 400 μL of 1% (vol/vol) EGDMA in HEMA, 100 μL of 6% (wt/vol) V-70, and 4 mL of methanol.
6 Transfer the mixture to the precut 1/8-inch-i.d. FEP tubes with a glass pipette, and secure the open ends of the tubes with microstoppers.
7 Place the tubes in a 50 °C oven overnight until polymerization is complete.
8 Cut the poly-HEMA gel into 2-cm segments with a razor blade while the gel is still inside the FEP tubing. The gel sticks to the tubing at this point.
9 Cure the gel at 65 °C in the incubator for 12 h; after curing, the gels can be pushed out easily with a 3-mm-o.d. NMR tube.
10 Wash and dry the gels by following the same steps described previously for the preparation of poly-HEMA gels for use with the compression device.

### Equipment setup

**Gel transfer funnel for the stretching device**

To make the transfer funnel, cut the narrow end of a 1-mL pipette tip with a razor blade (Fig. 2a). The opening at the cut end should have an i.d. of 4.2 mm to allow the Shigemi plunger to pass through smoothly (Fig. 2b). Connect the funnel to the wide end of the two-stage stretch NMR tube with a short length of rubber tubing (Fig. 2c). Also, prepare the microstoppers for both the 4.2-mm-i.d. and 3.2-mm-i.d. ends (Fig. 2d). ▲ CRITICAL Ensure that the cut end of the pipette tip has the same inner diameter as the wide end of the sample tube, i.e., 4.2 mm, so that the gel can be smoothly transferred through the funnel.

**Rubber stoppers for the stretching device**

To make the stoppers, cut the sharp end of a microstopper (the big piece on the left in Fig. 2d) with a razor blade into small cylindrical pieces (the two small pieces to the right in Fig. 2d) that fit into the stretching tube openings of different inner diameters, e.g., 4.2 and 3.2 mm.

## Procedure

▲ CRITICAL It is recommended that all isotropic solution NMR measurements needed for conventional structural analysis, such as 1D proton, carbon, COSY (homonuclear correlation spectroscopy), HSQC, HMBC, and nuclear Overhauser effect spectroscopy (NOESY)/rotational frame nuclear Overhauser effect spectroscopy (ROESY), be completed before using the deuterated solvent solution of the molecule to soak the gel in preparation for acquiring the anisotropic NMR data.

▲ CRITICAL In what follows, the term 'stick' is used to refer to a dry polymeric gel; the term 'gel' is used to refer to the material after it has been swollen with a deuterated solvent.

### Choose the alignment system ● Timing 10 min

1 Select an NMR solvent that yields the best spectral quality and sample stability. Such information should already be known from regular isotropic NMR studies. Next, choose an appropriate type of

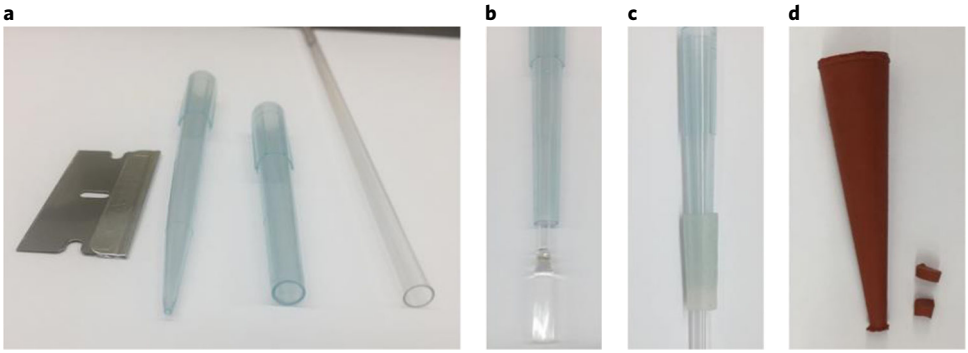



**Fig. 2 | Making the gel transfer funnel and the stopper for use with the stretching device. a**, A 1-mL pipette tip (the second item from the left) is cut with a razor blade (first item from the left) for use as the transfer funnel (the third item from the left), so that the opening has the same inner diameter as the wide end of the stretching device tube (the fourth item from the left), i.e., a 4.2-mm i.d. **b**, Ensure that the Shigemi plunger can smoothly pass through the funnel's opening. **c**, Connect the funnel to the 4.2-mm-i.d. end of the stretching device with rubber tubing. **d**, The microstopper from Fisher (left) is cut at the sharp end into two small pieces (right), which should fit the 4.2-mm-i.d. and 3.2-mm-i.d. ends of the stretching device.

**Table 1 | Comparison of gel compression and stretching methods**

| | Compression | Stretching |
|---|---|---|
| Type of anisotropic data | RDCs and RCSAs | RDCs and RCSAs |
| Commercial equipment available | Yes | Yes |
| Reversible alignment adjustment | Yes | No |
| Simultaneous *J* and RDC measurement | Yes | No |
| NMR spectral complexity | High (both anisotropic and isotropic) | Low (anisotropic only) |
| Isotropic shift correction for RCSAs | Needed | Not needed |
| Sample mass requirement | 1 mg or less | >3 mg for PMMA; less for poly-HEMA |
| NMR experiment time limit | No | <12 h for PMMA; none for poly-HEMA |

gel that is compatible with that solvent; refer to the introductory section 'Compressed or stretched polymeric gels as the alignment media for small organic molecules'.

2 Choose between gel compression or stretching as the alignment method. From a structure elucidation perspective, the compression and stretching methods are equivalent; both afford highly correlated data[38]. In practical terms, some differences exist, as was noted earlier in the 'Properties of the compression and the stretching methods' section. Differences are summarized in Table 1 to facilitate choosing the method that best suits the problem at hand.

## Gel preparation and quality control ● Timing 3 d

3 Follow our previously published recipes to make the chloroform-compatible PMMA gel or the DMSO-compatible poly-HEMA gel[38,63,64]. Refer to the Reagent setup section for the actual steps. The list of key components and reagents for the preparation of the gels is in Table 2. Once an investigator has developed proficiency in preparing the gels, it is useful to make several different types of gels and store them for future use. Note that several types of compression gels are now commercially available from Mestrelab Research, including PMMA (chloroform), poly-HEMA (DMSO), and poly-DEGMEMA (methanol).

▲ CRITICAL STEP The recipes for compression and stretching gels are different; the gels produced cannot be used interchangeably.

4 Check the quality of the gels before using them with the analyte. A quality control sample can be prepared following the same procedure as for the experimental sample, as described in Step 5, except

**Table 2 | Key components and reagents needed for PMMA and poly-HEMA gels**

| | PMMA | | Poly-HEMA | |
|---|---|---|---|---|
| | Compression | Stretching | Compression | Stretching |
| Monomer (vol/vol) | MMA (83.3%) | MMA (60–80%) | HEMA (83.3%) | HEMA (60%) |
| Cross-linker (mol/mol) | EDGMA/MMA (0.27%) | EDGMA/MMA (0.01–0.02%) | EDGMA/HEMA (0.2–0.3%) | EDGMA/HEMA (0.03–0.05%) |
| Initiator (wt/vol) | V-70 (0.025%) | AIBN (0.03%) | V-70 (0.025%) | V-70 (0.06%) |
| Solvent | Acetone | Acetone | Methanol | Methanol |
| Temperature | 50 °C | 65 °C (AIBN) | 50 °C | 50/65 °C[a] |
| Casting tube | 3-mm NMR tube | 4-mm NMR tube | Silanized 3-mm NMR tube | 1/8-inch-i.d. FEP tubing |
| Cut gel length | 2.5 cm | 2 cm | 2.5 cm | 2 cm |
| Prewash solvent | Acetone/methanol (1:1) | Acetone/methanol (1:1) | Not needed | Not needed |
| Prewash time | 12 h | 12 h | NA | NA |
| Wash solvent | Chloroform | Chloroform | Methanol | Methanol |
| Wash time | Three times over 1 d | Three times over 1 d | Three times over 2 d | Three times over 2 d |
| Gel drying time | 12 h in hood | 12 h in hood | 12 h in hood | 12 h in hood |
| Dry gel diameter | 2 mm | 3 mm | 2 mm | 3 mm |
| Solution volume | 450 μL | 350 μL | 450 μL | 330 μL |

NA, not applicable. [a]Polymerization at 50 °C and curing at 65 °C.

that the gel stick is swollen without adding any analyte. The gel quality can be assessed by examining the line shape and alignment of the $^2$H NMR spectrum. The figure in Box 2 shows what the typical $^2$H spectra should look like for compression and stretching devices. If your spectra do not look like those shown in Box 2, refer to the Troubleshooting section. The gel quality can also be examined in more detail using deuterium 1D imaging and MRI microimaging[85]. The quality check step is not needed for commercial gel products (Mestrelab Research).

**? TROUBLESHOOTING**

### Prepare the 'weakly' aligned gel sample for NMR measurement ● Timing 12–48 h

▲ **CRITICAL** Ensure that the appropriate solution volume is used to swell the gel; refer to Table 2.

5 Perform the steps in option A or B to make a weakly aligned NMR sample to analyze by the compression or stretching method, respectively. Note that all accessories, such as the transfer funnel and rubber stoppers prepared as part of option B, can be kept for long-term use.

(A) **NMR sample for the compression method**

(i) Insert a gel stick into the specialized NMR tube of the compression device (Fig. 3a) and add the analyte solution (450 μL) to the tube (Fig. 3b). This solution can optionally contain 5 μL of TMS for carbon chemical shift referencing to facilitate RCSA measurement. As the PMMA gel floats in $CDCl_3$, insert a piston to keep the gel fully submerged in the solvent (Fig. 3b).

(ii) For PMMA gels, allow the gel to swell in a vertical position for at least 5 h, until it reaches a length of ~40 mm (Fig. 3c); for poly-HEMA gels, the gel should be allowed to swell for a minimum of 5 h, although overnight is preferable. Unlike PMMA gels in chloroform, poly-HEMA gels do not float in DMSO. After the gel fully expands, some liquid should still be present in the tube.

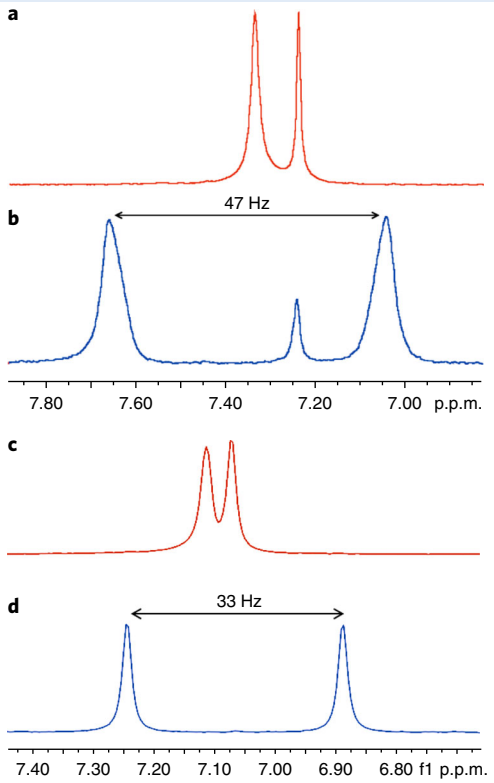
(iii) Compress the gel slightly by gently pressing down and locking the piston. There should be sufficient liquid in the tube so that no air bubbles are present beneath the piston after mild compression. Now the sample is ready for NMR measurement in the weakly aligned condition (Fig. 3d). After the NMR experiments, the gel will be fully compressed (Fig. 3e), as described in Step 10a.

(B) **NMR sample for the stretching method**

(i) Soak a gel stick in 350 μL of $CDCl_3$ sample solution, for PMMA, or 330 μL of DMSO-$d_6$ solution, for poly-HEMA. Optionally, 5 μL of TMS can be added for carbon chemical shift referencing when measuring RCSA values.

**Box 2 | Using residual $^2$H quadrupolar coupling (RQC) of the solvent as a control measure of gel quality**

The anisotropic environment inside the strained gel matrix aligns not only the analyte, but also the solvent molecules. Although the small solvent molecules align only very weakly, the large interaction between the electric quadrupole moment of the deuterium nucleus and the electric field gradient at the nucleus still generates a measurable coupling that is several tens of hertz in strained gels; this is known as residual quadrupolar coupling (RQC). Panel a in the figure shows the $^2$H spectrum of $CDCl_3$ in a minimally compressed PMMA gel. The broad and sharp peaks in this spectrum arise from chloroform molecules residing inside the gel matrix and in the interstitial space between the gel and the tube wall, respectively; with minimal compression, RQC may not be resolved in the broader peak; upon maximum compression, the broad peak splits into a doublet separated by 47 Hz, whereas the singlet retains the same line shape but weakens substantially due to shrinkage of the interstitial space under strong compression, as shown in panel b. The same behavior of the RQC doublet is observed for stretched PMMA gels, except that here the isotropic $CDCl_3$ signal is absent in both panels **c** and **d**, because there is no interstitial space in the stretched configuration. RQC is usually resolved even at the weak alignment condition, as shown in panel **c**. The quality of the poly-HEMA gel is evaluated from the $^2$H RQC of aligned DMSO-$d_6$ in a manner similar to that described for the PMMA gel.



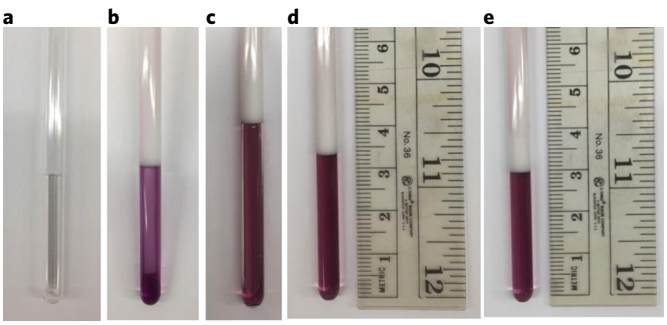




**Fig. 3 | NMR sample preparation with the compression device. a**, A PMMA gel 'stick' in the compression device. **b**, A solution of cryptolepine in 450 μL of $CDCl_3$ is added to the gel. The intense violet color of the alkaloid cryptolepine is used in our illustrations to facilitate the visualization of the gel. **c**, The fully swollen gel. **d**, The 'weakly' compressed gel. **e**, The 'strongly' compressed gel.

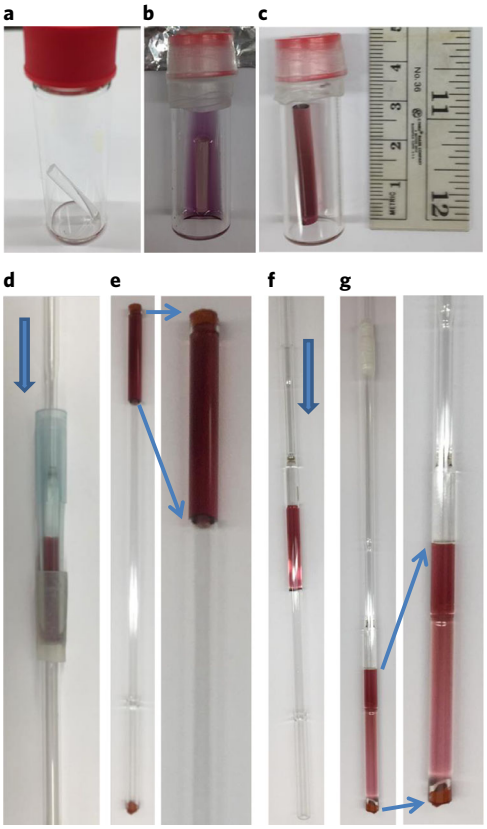



**Fig. 4 | NMR sample preparation with the stretching device. a**, A PMMA gel stick in a 2-mL glass vial. **b**, A solution of cryptolepine in 350 μL of $CDCl_3$ has been added. **c**, The fully swollen gel. **d**, The swollen gel is loaded into the transfer funnel, and a piston is used to push the gel into the stretch device. **e**, The 'weakly' aligned sample: the gel is fully transferred to the 4.2-mm-i.d. end, and two rubber plugs are inserted to seal the two openings. **f**, After NMR measurements for the weakly aligned sample, the gel is being pushed to the 3.2-mm-i.d. end. **g**, The 'strongly' aligned sample.

(ii) Perform gel swelling in a 2-mL flat-bottom vial placed in a nearly horizontal position but with the cap end slightly elevated. Swelling of the PMMA stick will complete overnight, whereas the poly-HEMA stick takes 2 d to fully equilibrate. See Fig. 4a–c. All liquid should be absorbed in the gel. The gel should have a diameter slightly larger than 4.2 mm, so no gaps should exist between the gel and the wall of the tube after the transfer.

(iii) Transfer the gel from the 2-mL vial to the funnel with a pair of tweezers, and gently push the entire gel through the funnel into the 4.2-mm-i.d. portion of the stretch tube, using a Shigemi piston (Fig. 4d), until the rear end of the gel is ~1 mm past the 4.2-mm-i.d. opening. Use a small amount of deuterated solvent to moisten the inner surfaces of the funnel and the sample tube just before the gel transfer, as the gel may be 'sticky'. This step should be completed within 1–2 min to minimize evaporation of $CDCl_3$.

**! CAUTION** As the Shigemi piston starts to contact the 4.2-mm-i.d. opening, push straight forward very slowly and carefully to avoid breaking the tube opening.

(iv) Seal the two openings of the sample tube with rubber plugs, which are cut to size from commercially available microstoppers (Fig. 2d). The rubber plugs should be cut for both sides to the smallest possible size that still allows good sealing, so that the gel fills the entire probe rf coil without the stopper intruding into the coil. Now NMR measurements for the weakly aligned sample can be performed with the gel in the 4.2-mm-i.d. section of the tube (Fig. 4e). After the NMR experiments, the gel will be translocated to the narrow end of the stretching device with the Shigemi piston (Fig. 4f), where the gel will enlongate under radial constriction (Fig. 4g), as described in Step 10b.

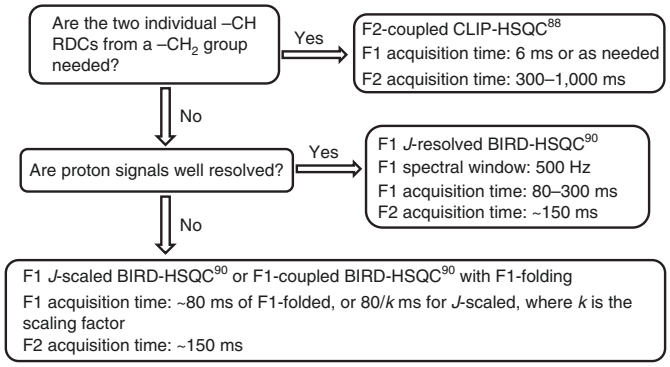



**Fig. 5 |** Flowchart for choosing the best pulse sequence for measuring the total one-bond $^{1}$H–$^{13}$C *T* couplings.

### Perform NMR measurement of the weakly aligned sample ● Timing 2–12 h

6 Shim the sample. Automatic shimming occasionally fails for gel samples. Even when automatic shimming succeeds, it is still recommended to refine the shimming by manual shimming on free induction decay (FID). One can either shim for the best linewidth of TMS (if present) or the residual solvent proton peak. A few cycles of shim adjustments in the order of $z$–$z^2$–$z^3$–$z^4$–$x$–$y$–$xz$–$yz$ usually yield satisfactory results. For the compression method, it is usually a good idea to perform automatic shimming for an identical tube/device, adjusting the position of the piston to be at the same height as in the tube containing the gel. After that, minor adjustment of $z1$ and $z2$ shimming usually suffices. The shim values can be saved and reloaded in later experiments as a good starting point. The TMS peak usually has a half-height linewidth of 2–3 Hz for PMMA gels and 3–5 Hz for poly-HEMA gels after shimming. Poor shimming is typically associated with broad and/or asymmetric line shapes.

7 Collect a $^{2}$H spectrum for the deuterated solvent. As the gel is only weakly constrained, the residual $^{2}$H quadrupolar coupling (RQC) is usually small or unresolved. For a PMMA gel in a stretching device, an RQC of ~3–10 Hz can usually be observed at this step. For a PMMA gel in a compression device, or a poly-HEMA gel in either type of device, the RQC is typically unresolved.

8 Collect a 1D $^{13}$C{$^{1}$H} NMR spectrum. Typical acquisition parameter settings used for regular isotropic samples can be applied. While processing the data, apply a generous zero-filling, e.g., to 1,024 K points. An exponential line broadening of 1–2 Hz can be applied.

9 Collect the total $^{1}$H–$^{13}$C couplings (*T*) using an F2-coupled CLIP–HSQC experiment or one of the variants of F1-coupled HSQC, or both. As described in the 'Experimental measurement of anisotropic NMR data' section, the total couplings can be measured from two types of experiments that differ in whether the couplings are resolved in the F2 (proton) dimension or F1 (carbon) dimension. In general, the measurement accuracy is higher with the second type because of a sharper line shape, whereas the first type provides more information by yielding RDCs of individual CH bond vectors in a methylene group. See Fig. 5 for advice on the decision-making process.

### Prepare the 'strongly' aligned gel sample for NMR measurement ● Timing 5 min

10 Remove the sample from the magnet and create the 'strong' alignment condition using the procedures described below for the compression (option A) or stretching method (option B).

(A) **Compression**

(i) Unlock the piston and gently push it straight down, until the gel is compressed to ~30 mm in length.

▲ CRITICAL STEP Do not adjust the piston position by turning the knob on the compression device. This practice may twist and break the gel.

? TROUBLESHOOTING

(ii) Relock the piston (Fig. 3e).

(B) **Stretching**

(i) Unplug both ends of the tube by removing the rubber stoppers, and then use a Shigemi piston to gently push the gel toward the narrow (e.g., 3.2-mm i.d.) opening (Fig. 4f), until the front end of the gel is ~1 mm from the mouth of the narrow opening.

? TROUBLESHOOTING

(ii) Reseal the narrow opening with the rubber stopper with gentle pressure on the piston to prevent the gel from backsliding, and then secure the piston using Parafilm wraps to close the wide (4.2-mm-i.d.) opening (Fig. 4g). The NMR measurement in the strongly aligned condition will be performed with the 3.2-mm-i.d. section of the tube positioned in the rf coil (the tube is inverted relative to the weakly aligned condition).

### Perform NMR measurement for the strongly aligned sample ● Timing 2–36 h

11 Shim the sample as in Step 6, beginning from the reloaded shims. The achieved linewidth should be similar to that achieved in Step 6.

12 Collect a $^{2}$H spectrum to evaluate the degree of alignment from the RQC of the deuterated solvent. For the stretched PMMA gel, the $CDCl_3$ RQC can typically range from 30 to 50 Hz in a 3.2-mm-i.d. tube; similar values are expected from the compression device as well. For a poly-HEMA gel with 0.04–0.05% (mol/mol) cross-linking (Table 2), the DMSO-$d_6$ RQC is typically 6–8 Hz in a 3.2-mm-i.d. tube.

13 Collect a 1D $^{13}$C{$^{1}$H} NMR spectrum. Regular parameter settings for isotropic measurements can be used. One can align this spectrum with the one collected at Step 8 for a quick visual check of the RCSA data. If the stretching method is used and TMS is added for carbon chemical shift referencing, aligning the two carbon spectra by superimposing the TMS resonances can directly report the RCSA of each carbon as the offset between the corresponding peaks in the two spectra. An RCSA has a positive sign if the carbon resonance under the strong alignment shifts downfield relative to the weak alignment, or a negative sign if the opposite scenario is the case. If the compression method is used, the RCSA cannot be directly interpreted; refer to the Supplementary Methods for procedures for analyzing RCSA data collected in compressed gels.

▲ CRITICAL STEP The $sp^2$ carbons usually exhibit larger RCSAs than do $sp^3$ carbons, with maximal magnitudes typically ranging from 5 to 10 Hz on a 600-MHz instrument, depending on the alignment propensity of the analyte, as well as the gel used and the degree of strain applied. The $sp^3$ carbons generally show RCSAs a few times smaller than those of $sp^2$ carbons, but note that $sp^2$ carbons do not always yield larger RCSAs than do $sp^3$ carbons, as the CSA tensor orientation relative to the alignment tensor also affects the size of the RCSA. If all RCSA values in an $sp^2$ carbon–rich compound are <3 Hz, one should consider increasing the alignment amplitude based on the information provided in the 'Adjust the alignment amplitude if needed' section. However, if the compound is deficient in $sp^2$ carbons, small RCSA values do not necessarily indicate weak alignment, and in this case, RDC data should be collected as well before deciding whether the alignment strength should be increased.

14 Collect the total $^{1}$H–$^{13}$C $T$ couplings with the same pulse sequences as those used in Step 9. Superimpose the resulting spectra with the corresponding spectra from Step 9 to evaluate the RDC data. The methods described in this protocol should generate $^{1}$H–$^{13}$C RDCs smaller in magnitude than those of $^{1}J_{HC}$ couplings, and therefore an RDC has a positive sign if the total coupling increases in the strongly aligned sample, or a negative sign in the opposite case.

### Adjust the alignment amplitude if needed ● Timing 10 min–12 h

15 Examine the RDC values. The maximal one-bond $^{1}$H–$^{13}$C RDCs in a molecule are typically between 10 and 30 Hz in magnitude. If the largest one-bond $^{1}$H–$^{13}$C RDC measured is <10 Hz in magnitude, the alignment is usually considered too weak. Conversely, when a one-bond $^{1}$H–$^{13}$C RDC is >50 Hz, the corresponding signals in the HSQC-based experiments can be very weak, for the reasons cited in the 'Tuning the alignment' section. Use the guidelines provided in Box 3 and data available from the first trial to determine the optimal alignment conditions and then reacquire the NMR data, if needed.

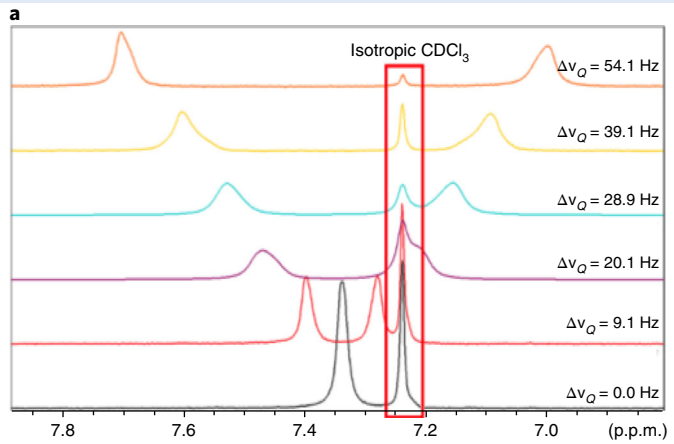
### Computation of 3D conformations and chemical shielding tensors ● Timing variable, ranging from a few hours to 1 d

▲ CRITICAL These steps can be performed in parallel with NMR sample preparation and NMR data acquisition. The requisite molecular modeling is essentially identical to that previously reported by Willoughby, Jansma, and Hoyce[107], and their Python scripts, intended to work within the Maestro environment, can be easily adapted to the present protocol by just changing the combination of functionals and basis sets.

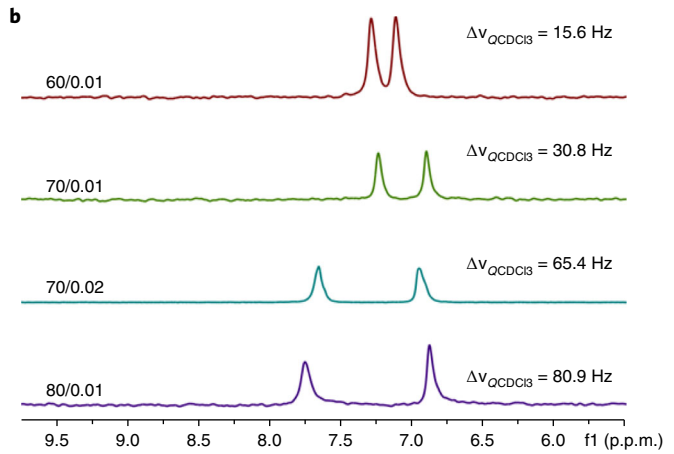
16 Draw all structural candidates to be analyzed in ChemDraw. The stereochemistry must be defined for all candidates. As the current method can determine only relative stereochemistry, only one of the two enantiomers needs to be specified for each candidate. This may take 10–20 min.

**Box 3 | Adjusting the alignment amplitude**

In the compression device, the alignment amplitude can be controlled by fixing the position of the plunger (piston) at different levels, thereby affording differing degrees of gel compression (panel **a** in the figure).
In the stretching device, two approaches can be used. First, one can choose different inner diameter ratios for the wide versus narrow sections of the stretching device. A standard design is a 4.2-mm i.d. and a 3.2-mm i.d. for the wide and narrow tube sections, respectively, which is referred to as 4.2/3.2 hereafter. A smaller inner diameter for the narrow 60/0.01 section, for example, 3.0 mm, yields an alignment that is ~40% stronger.
Another approach to alignment modulation is using a gel polymerized with different monomer concentrations or cross-linking ratios (panel **b** in the figure). A higher monomer concentration or cross-linking ratio steeply increases the alignment amplitude. A PMMA gel polymerized from a methyl methacrylate concentration of 70% (vol/vol) and a cross-linking ratio of 0.01% (mol/mol), referred to as PMMA 70/0.01 hereafter, when used with a 4.2/3.2 stretch tube, generally provides a satisfactory RDC range for most small molecules. For a molecule that aligns too weakly, PMMA 70/0.02 can be used with a 4.2/3.2 tube for a doubling of alignment amplitude; conversely, for a molecule that aligns too strongly, PMMA 70/0.01 can be used with a 4.2/3.4 tube for a ~50% reduction of alignment, or PMMA 60/0.01 can be used with a 4.2/3.2 tube for a similar reduction of alignment.




1D $^2$H NMR spectra of $CDCl_3$-swollen PMMA gel at different degrees of compression.




1D $^2$H NMR spectra of $CDCl_3$ swollen PMMA gel polymerized with different monomer concentrations or cross-linking ratios, stretched with a 4.2-3.2-mm NMR tube.

17 Export the ChemDraw.cdx file to ChemDraw 3D, perform a minimization with MMFF, and save the coordinates in a.sdf file. Alternatively, the multiplatform Open Babel program[108] can be employed using the –gen3d option. This may take 5 min.

18 Generate a project file in Maestro and import the.sdf file generated in Step 17. Verify that configuration did not change during the generation of 3D coordinates. Call the Macromodel program and select the MCMM conformational method. A few to hundreds of conformers may be generated, depending on the flexibility of the input structure. This takes <1 h.

```
rdc_data {
#c1 h1
17 48 17.0
#c6 h6
16 47 2.5
#c5 h
15 46 -23.8
……
#NMe 31
18 49 -6.6
18 50 -6.6
18 51 -6.6
#NMe 32
31 53 -6.0
31 54 -6.0
31 55 -6.0
}

rcsa_data {
#c18
19 -0.014
#c11
5 0.0084
#c33
29 -0.0156
…

}
```

```
!* MSpin-RDC Plugin *!

********
!* Computation flags *!
Method: SVD
Scaling mode: Hz
Scale QCSA with axial component: False
Include CSA gel shift (isotropic) correction: False
Optimize CSA gel shift (isotropic) correction scale: False
Gel Shift Correction: 0.15
Single Tensor: True
Optimize populations: False
Grid search points: 16
Average methyl groups: True
Average methylene groups: False
Average phenyl groups: False
Bootstrapping: False
RDC Std. Error [ppm]: 1
CSA Std. Error [ppm]: 0.01
PCS Std. Error [ppm]: 0.01
DQ Std. Error [Hz]: 1
********
!* Permutations *!
There are no permutations on the original data set
********
Data set: #1
Computed data for frame #1
RDC Data:
I      J      Exp. [Hz]   Comp. [Hz]

C17    H48    17.00       18.31
C16    H47    2.50        2.93
C3     H42    24.50       22.01
C20    H52    3.30        2.93
C35    H59    26.50       28.29
……
```

**Fig. 6 | Excerpt from the Mspin input and output files for cryptospirolepine (2).** See complete examples in the Supplementary Information.

19 Export the Maestro output ensemble as a.sdf file to Gaussian 09 for geometry optimization and free-energy calculation. Use the Gaussian 09/16 keywords: `#B3LYP/6-31G** opt freq`.

20 Remove redundant structures from the Gaussian 09 output and discard any structure presenting negative vibrational frequencies. Calculate the Boltzmann populations for the remaining conformers based on the sum of electronic energies and thermal free energies. Depending on the structural flexibility, from one to tens of conformers may form the lowest-energy ensemble with a 2 kcal/mol cutoff.

21 Calculate chemical shielding tensors for all conformers with energies within 2 kcal/mol of the lowest one. Use the Gaussian 09/16 keywords: `#MPW1PW91/6-31G** nmr`.

## Anisotropic NMR data analysis by MSpin ● Timing 30–60 min

▲ **CRITICAL** The different options in MSpin for fitting RDC and RCSA data can be introduced either step by step through the graphical interface, or by the use of appropriate control keywords in the input file. In the following, we illustrate the recommended steps in this protocol. The steps below constitute a practical workflow.

22 *Create an RDC/RCSA input file*. See the entry for the cryptospirolepine *N*-Me groups in Fig. 6. The left column is the user input, whereas the right column shows the MSpin output. Make certain that the atomic numbering in the input file matches that of the Gaussian 09 output. Hint: MSpin allows the direct use of $^1D_{CH}$ couplings from $-CH_3$ and symmetrical phenyl groups. The procedure has been described by Sánchez-Pedregal et al.[97]. All equivalent coupled pairs must be explicitly written.

Note that if data from methylene groups are obtained from an experiment that yields only the sum of the two C–H couplings, the data should be entered into the input file as the half-sum of the couplings, i.e., the average value of the two C–H couplings.

23 Store all Gaussian 09 output files in the same directory.

24 *Load all conformations into MSpin*. This operation is achieved through the MSpin option *File → Open Directory*. At this point, all conformations are loaded and the chemical shielding tensors are stored in memory. Note that information about the tensors, such as the principal axis orientation, can be visualized through the CST plug-in of MSpin.

25 *Calculate Boltzmann populations*. By default, MSpin will assign equal populations to all conformations. Because SCF (self-consistent field) DFT energies are retrieved from the Gaussian 09 files in MSpin, Boltzmann populations can be computed. To do this, select the NVT ensemble (number of particles, volume, and temperature) and target temperature in the *Conformers* window. Alternatively, this can be requested through the `compute_populations {true}` control keyword. Note that computing Boltzmann populations from zero-point vibrationally corrected energies or Gibbs free energies would be a more accurate procedure. The user must separately tabulate these energies, compute the Boltzmann populations, and introduce them into the program, either through the *Paste population* option in the conformer window or by using the `populations` control keyword.

26 *Conformer superposition*. If more than one conformer is present and the single-tensor SVD approach is to be used, rotate the molecules to maximize superimposition of the atom coordinates in order to define a common frame.

27 (Recommended) Superimpose atom coordinates for all heavy atoms in the molecule. Go to *Window → Command Line*. An editable line is shown in the bottom of the window, and the following command must be entered: `world.eckarttransform()`. The coordinates of all non-hydrogen atoms are superimposed using a quaternion-based algorithm using mass-weighted coordinates. Note that an explicit list of atoms can be used instead. At this point, not only molecular geometries but also shielding tensors have been rotated to the new frame. Note that you can visualize the superimposition by using the command `world.showallframes(true)`). The superposition procedure can be requested from the input file using the control keywords `superimpose {true}`, `superimpose_mode {massweighted}`, and `superimpose_atoms {heavy}`.

28 Load the RDC/RCSA input file.

29 (Optional) Check that all RDCs and RCSAs are correctly assigned by checking the *Show RDCs* and *Show RCSAs* boxes, as well as the *View → Show Atom* indexes. For RCSA data collected from compressed gels, click the *Apply* and the *Estimate correction* checkboxes in the *Gel Shift correction* box in order to correct for isotropic chemical shift changes (see above). For RCSA data collected from stretched gels, proceed to the next step.

30 *Set SVD options*. Note that to request simultaneous optimization of the gel shift correction parameter, check the *Optimize* option. Go to the *computation* tab. Select the *SVD* option, as well as the *scaling in Hz* option. If more than one conformer is present, check the *single tensor* option (keyword: `single_tensor {true}`). (Optional) Optimize population amplitudes by checking the *fit populations* option (`optimize_populations {true}`). If $^1D_{CH}$ data from methyl groups or symmetrical phenyl groups are present (`average_methyl_groups {true}`; `average_phenyl_groups {true}`), check the respective options in the *Groups* box inside the *computation* tab). Important: check the *methylene* option if data from $CH_2$ groups were entered as the half-sum (`average_methylene_groups {true}`). Note that MSpin-2.4 and StereoFitter programs share a more general way to handle chemical equivalence. See the Supplementary Methods for further information and examples.

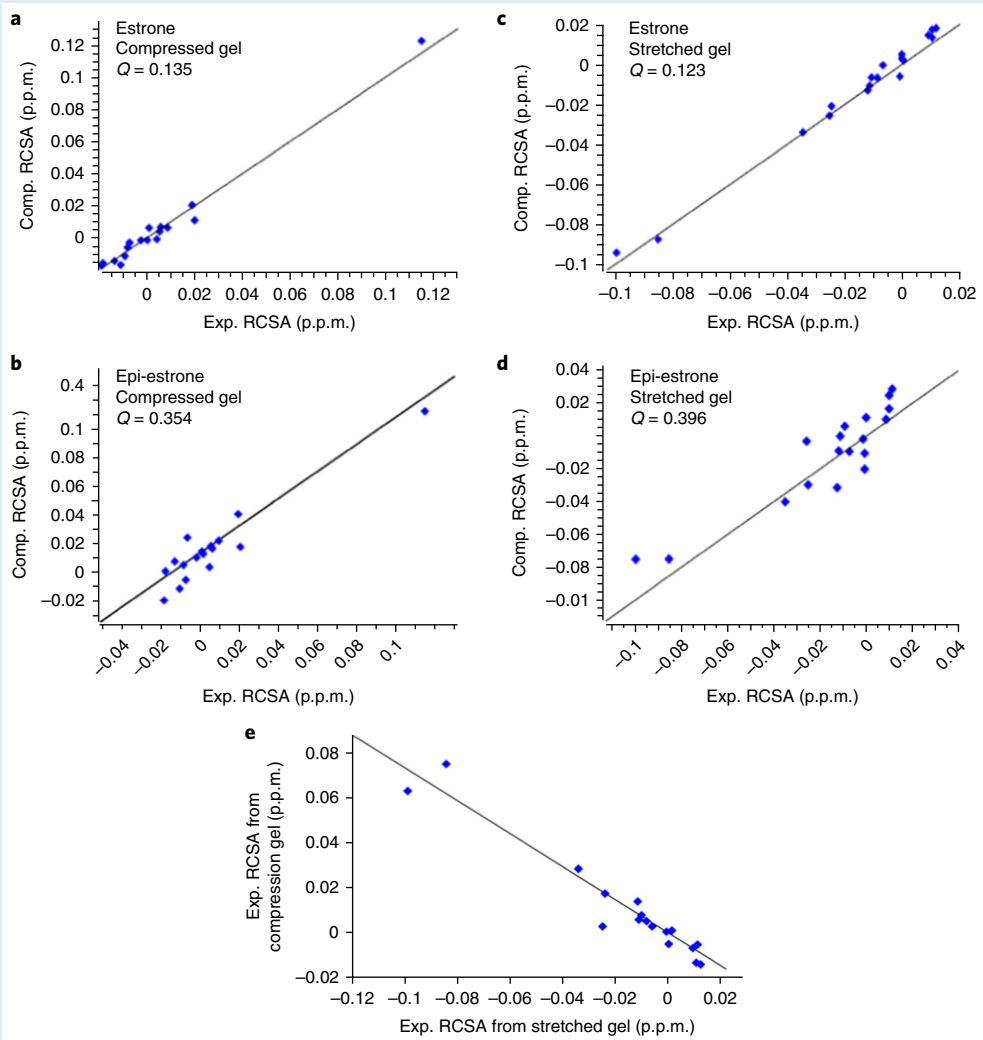
31 *Perform SVD*. Click on the *Calculate* button. Generally, the alignment tensor computation should finish within seconds, unless fitting of conformer populations is required, in which case the duration of the calculation can range from minutes to hours. Click on the *Show Results* button. Tables with experimental versus computed data are then shown, as well as the quality factor ($Q$) for the fitting.

**? TROUBLESHOOTING**

32 *Inspect SVD results*. The quality of the results can be further analyzed using several tools. Details on the SVD fitting and alignment tensor can be displayed by clicking on the *Show Results* button. Experimental versus back-calculated RDC or RCSA values are displayed in tabular form. Right-clicking on the table contents allows *xy* plots to be displayed by selecting the *Open XY Graph* menu

**Box 4 | Stereochemical differentiation of estrone (5) and 13-epi-estrone (6) by RCSA data in compressed and stretched gels**

The figure shows the correlation plots between experimental and back-calculated RCSAs generated by MSpin that allowed clear differentiation of estrone (**5**) from 13-epi-estrone (**6**). Panels **a** and **c** display good linearity and low *Q* factors for the correct structure **5**, which was analyzed with RCSA data collected in compressed and stretched gels, respectively. By contrast, panels **b** and **d** display much more scattered correlations and much higher *Q* factors for the incorrect structure **6**, which was analyzed using the same RCSA data. Such contrast in correlations and *Q* factors between candidates of different stereochemistries or constitutions forms the basis of structural differentiation by anisotropic NMR data. In panels **a** and **b**, experimental RCSA data from compression gel were the original uncorrected data, whereas the back-calculated data were subject to isotropic correction, as described in the Supplementary Information. Panel **e** shows the anti-correlation between the data from the stretched gel and data from the compression gel after isotropic correction.




option. In Box 4, we show an example of the *xy* plots produced by MSpin in which the anti-correlation between data acquired in stretched and compressed gels can be clearly appreciated. Graphical analysis of the alignment tensor can be conducted with several tools, such as graphical 2D or 3D representation of the principal axes, as well as a valued surface. Complete text output can be obtained using the *Show Results → Export Output* option. Note that you can visualize the alignment tensor properties by displaying principal axis frames or adding a valued surface constructed from the alignment tensor (Fig. 7). Select the *Display* tab on the RDC MSpin module and click on the *show* option in the *Alignment tensor 4D Surface* box. Axially symmetrical tensors will look like a $d_{z2}$ orbital, whereas maximally asymmetric tensors will look like a $d_{x2-y2}$ orbital. The

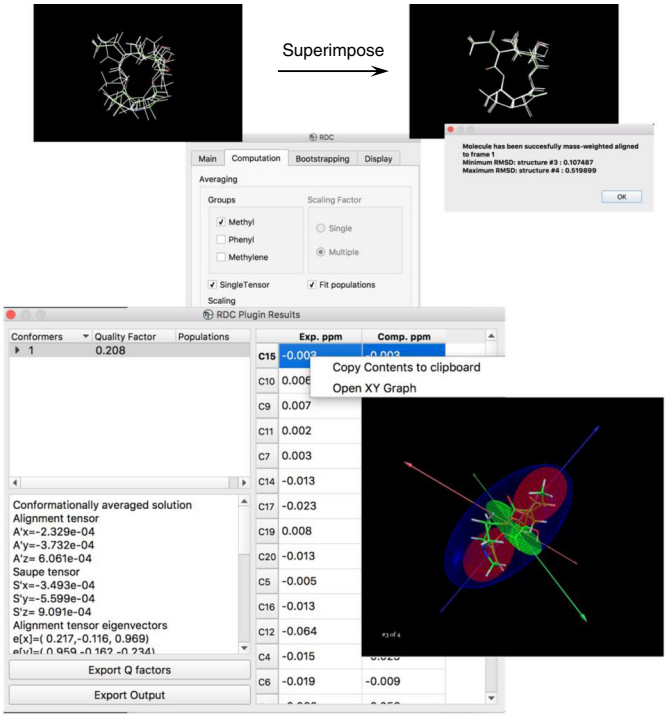



**Fig. 7 | Single-tensor RCSA computation for retrorsine.** Top left, original non-superimposed structures. Top right, structures after atom superimposition (Eckart transform). Middle, MSpin computation options. Bottom, MSpin results window and graphical display of valued surfaces (red, green), gyration ellipsoid (blue), and alignment tensor principal axes (blue, green, and red for *x*, *y*, and *z* axes, respectively).

distance from the centroid to the surface will be proportional to the RDC value pointing in that direction, whereas the color informs on the sign (negative $^1D_{CH}$ RDCs are colored red).

**? TROUBLESHOOTING**

33 *Repeat Steps 22–32, for all the structural proposals differing in configuration or constitution.* The *Q* values obtained for the different structures are then compared. The correct structure should exhibit a *Q* value typically<0.2 with good data quality. Structural differentiation is possible if the other structural proposals have substantially higher *Q* factors. (Optional) It is possible to perform bootstrapping computations in which Gaussian noise is added to each particular datum according to the estimated uncertainties. A distribution of *Q* factors is then computed from the generated decoy datasets. An s.e. of 1.0 Hz is a reasonable value, but it is convenient to repeat the bootstrapping computations at lower degrees of confidence. The structural verification workflow is summarized in Fig. 8.

## Sample recovery ● Timing 1–2 d

**▲CRITICAL** Samples from both compressed and stretched gels can be almost completely recovered. Sample recovery is typically >90% according to quantitative NMR.

34 To extract the gel from the compression device, insert the sample tube open end down in a 10-mL centrifuge tube (~100*g*, ~25 °C, 20 s) and spin the gel out, along with any liquid, using a hand centrifuge.

35 The gel in the stretching device can simply be pushed out through the narrow open end. Perform this step using a thin glass rod of ~3-mm diameter, going in from the wide end.

36 For PMMA gels, extract the analyte from the gel through diffusion with 3 mL of chloroform with an extraction time of 8 h. This process should be repeated four times to ensure complete sample recovery. The gel formulated for stretching grows to a much larger size during extraction. It is advisable to cut the gel in half to ensure that each piece is completely submerged in chloroform for efficient extraction as the gel expands.

37 Extraction from the poly-HEMA gel is similar. Use DMSO as the solvent and extract for 12 h for each of the four extraction intervals.

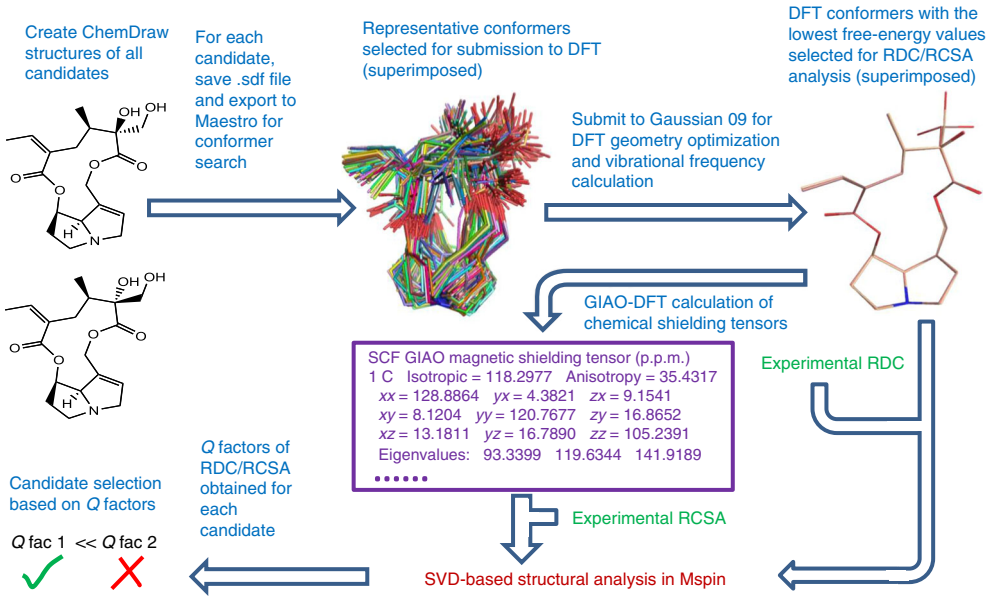



**Fig. 8 | Flowchart of the structural verification process using retrorsine as an example.** This process contains three stages: candidate generation (top left), RDC/RCSA-based structural analysis (right), and *Q*-factor-based candidate selection (bottom left). Note that during candidate selection, the lowest *Q* factor represents the 'best' candidate rather than the 'correct' candidate; it is critical to have the correct structure in the evaluation pool in order to avoid false-positive structural identification.

## Troubleshooting

### The gel breaks when I compress it (Step 10A(i))

- Do not adjust the piston position by turning the knob on the compression device. This practice may twist and break the gel. Instead, unlock the piston first, move the piston straight up or down, and relock it. When unlocking the piston from the compressed state, apply gentle pressure to prevent sudden upward movement, which can cause suction and break the top of the gel.

### The poly-HEMA breaks when I stretch it (Steps 4 and 11B(i))

The length of the dry stick and the solution volume used to swell it should be matched, so that the gel diameter is ~4.2 mm after swelling. Refer to Table 2 for details. Make certain that the entire gel is in contact with the liquid in a horizontal position during swelling, so that a uniform cylinder is obtained after the gel fully swells. Allow the gel to fully equilibrate for at least 2 d. In the case that the bottom of the gel breaks during stretching, use the piston to push the broken pieces very slowly out of the narrow opening; use a pair of tweezers to remove the broken pieces, if necessary. The remaining intact part of the gel can be used for NMR data collection, provided that the gel passes the quality control test, as described in Step 4.

### I am getting a poor *Q* factor (Steps 31 and 32)

It is always a good idea to check the pairwise correlation between experimental data and back-calculated data from MSpin, in addition to the *Q* factor (Box 4). This correlation can be visualized through a correlation plot with experimental data on the *x* axis and the back-calculated data on the *y* axis. If outliers are observed, one should look for factors that potentially affect these data, such as spectral data quality (either low signal-to-noise ratio or strong coupling effects), possible atom number mismatch in the data input, and structural flexibility. If structural flexibility is a potential cause, single-tensor SVD with an ensemble of low-energy rotamers can be applied, as described in Steps 24–26. If the correlation is poor overall, the molecule may have flexibility that substantially changes the molecular shape; such flexibility should be apparent from DFT geometry optimization. Of course, a poor *Q* factor or correlation may also indicate an incorrect structural proposal[39].

### RDCs and RCSAs fit well individually, but not when used together (Step 32)

It is important to ensure that the chemical shielding tensors used for RCSA analysis are calculated based on the same 3D coordinate that is used for RDC analysis. If the coordinate is rotated through software, the chemical shielding tensors must also be rotated in the same manner. This is ensured if MSpin is used for the analysis, as it reads both chemical shielding tensors and the coordinate from the same Gaussian 09 output file. Also, make sure that the signs of the RCSAs and RDCs are both correctly determined; see Steps 13 and 14.

### I obtain similar *Q* factors for different structure proposals (Step 32)

In general, the lack of differentiation between different structure proposals suggests that the data acquired are insufficient to accurately describe the molecule being studied, although the analysis method used may have a substantial effect as well. If only RDC or RCSA data are available, collecting the missing data should provide additional resolving power. In addition to the most frequently acquired one-bond $^{1}H–^{13}C$ RDCs, long-range RDCs can also be used, although the measurement is more challenging[109,110]. If the compound contains other NMR-active isotopes, such as $^{19}F$ or $^{31}P$[28,111], their RDCs to nearby carbons or protons are usually easy to measure using 1D $^{13}C$ or $^{1}H$ spectra and can be used for analysis as well. When RCSA data are used, enhanced differentiation can often be obtained through the use of the QCSA factor (i.e., a *Q* factor with residual chemical shift anisotropy weighting) in lieu of the conventional *Q* factor[38]. The QCSA factor equalizes the contributions of $sp^2$ and $sp^3$ hybridized carbons to structural differentiation, which is useful for a compound having a large number of $sp^2$ carbons but with the portion of the structure to be determined predominantly composed of $sp^3$ carbons. For example, non-distinguishing *Q* factors of 0.040 and 0.046 were obtained for RCSA data collected in stretched PMMA gels for erythro-mefloquine and threo-mefloquine, respectively, whereas the corresponding QCSA factors of 0.290 and 0.650, respectively, for the two molecules were clearly more differentiating[38]. Finally, orientating a small molecule in different alignment media can often produce orthogonal data and thereby increase the chances of discriminating between the different proposals being considered, as discussed in the 'Anisotropic NMR data analysis' section.

## Timing

Steps 1 and 2, choose the alignment system: 10 min
Steps 3 and 4, gel preparation and quality control: 3 d
Step 5, prepare the weakly aligned gel sample for NMR measurement: 12–48 h
Steps 6–9, perform NMR measurement of the weakly aligned sample: 2–12 h
Step 10, prepare the strongly aligned gel sample for NMR measurement: 5 min
Steps 11–14, perform NMR measurement for the strongly aligned sample: 2–36 h
Step 15, adjust the alignment amplitude if needed: 10 min–12 h
Steps 16–21, computation of 3D conformations and chemical shielding tensors: variable, mainly depending on how long it takes to perform the DFT geometry optimization for all conformers; depending on the complexity and flexibility, optimization of all conformers can take from a few hours up to a full day; calculation of the shielding tensors typically takes 10–30 min
Steps 22–33, anisotropic NMR data analysis by MSpin: 30–60 min
Steps 34–37, sample recovery: 1–2 d

## Anticipated results

Gel swelling, compression, and stretching should produce similar results, as demonstrated in Figs. 3 and 4. The gel sample should pass quality control by $^{2}H$ NMR, as described in Box 2 and Steps 8 and 13. Sample shimming should be performed so that TMS or solvent line shape meets the specifications in Steps 6 and 11. With a suitable degree of alignment, RCSA and RDC data should fall into ranges as specified in Steps 13 and 15. If the alignment is too weak or too strong, a desired degree of alignment should be achievable based on the guidance provided in Box 3 and Step 15. If the proposed structure is correct and computational modeling is properly performed, a low *Q* factor and good correlation linearity between measured and back-calculated anisotropic NMR values should be obtained, whereas incorrect structures will be associated with high *Q* factors and poor linearities, as illustrated in Box 4.

### Reporting Summary

Further information on research design is available in the Nature Research Reporting Summary linked to this paper.

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## Author contributions

G.E.M. and R.T.W. conceived the idea of writing this protocol; Y.L. and A.N.-V. contributed equally to manuscript writing. G.E.M. contributed the introduction and proposed the organization of the manuscript. R.R.G. contributed backgrounds and methods in gel preparation and gel compression. C.G. contributed theoretical background. All authors contributed to manuscript editing. G.E.M. coordinated the writing of the manuscript.

## Competing interests

A.N.-V. is the author of MSpin and StereoFitter, which are mentioned in the article, and receives royalties from sales of these products. The other authors declare no competing interests.

## Additional information

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